



MODERN INSTITUTE OF TECHNOLOGY & RESEARCH CENTRE, ALWAR

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Subject & Subject Code: BEE, 2FY2-07

Semester: II

Session: 2022-23 (Even Sem)

Branch: First Year

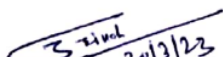
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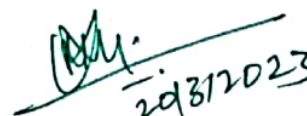
Unit: II

Date of Submission: 20/03/23

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Faculty


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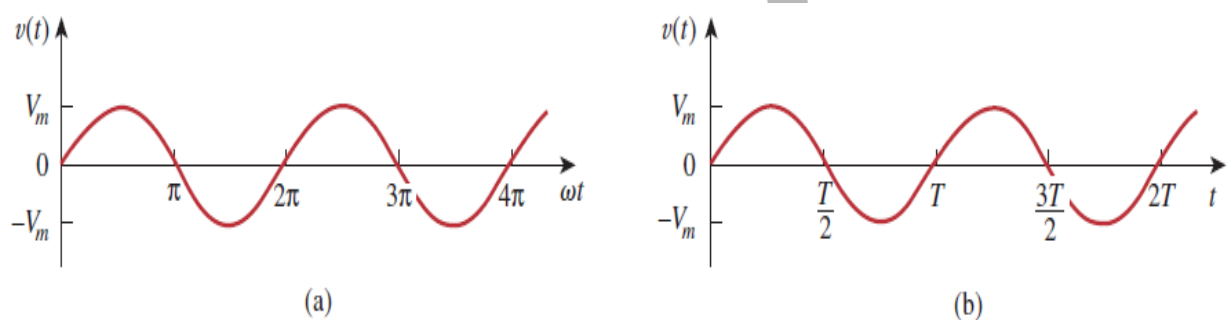
UNIT-2 (AC Circuits)

LECTURE NO:-1

Representation of Sinusoidal Waveforms, Peak and R.M.S Values

Representation of sinusoidal waveforms:

A sinusoid is a signal that has the form of the sine or cosine function. A sinusoidal current is usually referred to as alternating current (ac). Such a current reverses at regular time intervals and has alternately positive and negative values. Circuits driven by sinusoidal current or voltage sources are called ac circuits.



Consider the sinusoidal voltage

$$v(t) = V_m \sin \omega t$$

Where

V_m = the amplitude of the sinusoid

ω = the angular frequency in radians/s

t = time in second

The sinusoid is shown in Fig. (a) as a function of its argument and in Fig. (b) as a function of time. It is evident that the sinusoid repeats itself every T seconds; thus, T is called the period of the sinusoid.

From the two plots in Fig, we observe that

$$T = \frac{2\pi}{\omega}$$

Cycle

One complete set of positive and negative values of alternating quantity is known as cycle. A cycle may also be sometimes specified in terms of angular measure. In that case, one complete cycle is said to spread over 360° or 2π radians.

Time Period

The time taken by an alternating quantity to complete one cycle is called its time period T . For example, a 50-Hz alternating current has a time period of $1/50$ second.

Frequency

The number of cycles/second is called the frequency of the alternating quantity. Its unit is hertz (Hz).

It may be noted that the frequency is given by the reciprocal of the time period of the alternating quantity.

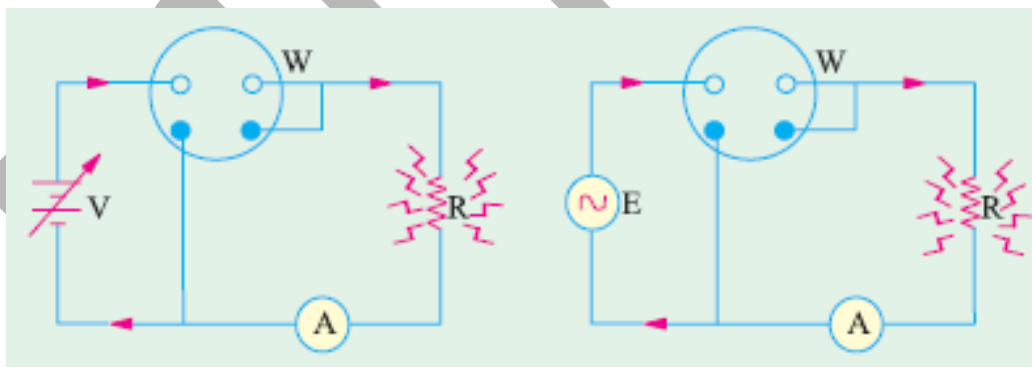
$$f = 1/T \text{ or } T = 1/f$$

Amplitude

The maximum value, positive or negative, of an alternating quantity is known as its amplitude.

Root-Mean-Square (R.M.S.) Value

The r.m.s. value of an alternating current is given by that steady (d.c.) current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time. It is also known as the effective or virtual value of the alternating current.



The standard form of a sinusoidal alternating current is

$$i = I_m \sin \omega t = I_m \sin \theta.$$

The mean of the squares of the instantaneous values of current over one complete cycle is (even the value over half a cycle will do).

$$= \int_0^{2\pi} \frac{i^2 d\theta}{(2\pi - 0)}$$

The square root of this value is $= \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi} \right)}$

Hence, the r.m.s. value of the alternating current is

$$I = \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi} \right)} = \sqrt{\left(\frac{I_m^2}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta \right)} \quad (\text{put } i = I_m \sin \theta)$$

$$\text{Now, } \cos 2\theta = 1 - 2 \sin^2 \theta \quad \therefore \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\begin{aligned} \therefore I &= \sqrt{\left(\frac{I_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta) d\theta \right)} = \sqrt{\left(\frac{I_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} \right)} \\ &= \sqrt{\frac{I_m^2}{4} \cdot 2} = \sqrt{\frac{I_m^2}{2}} \quad \therefore I = \frac{I_m}{\sqrt{2}} = 0.707 I_m \end{aligned}$$

Hence, we find that for a symmetrical sinusoidal current

r.m.s. value of current = 0.707 * max. value of current

- The r.m.s. value of an alternating current is of considerable importance in practice, because the ammeters and voltmeters record the r.m.s. value of alternating current and voltage respectively.
- In electrical engineering work, unless indicated otherwise, the values of the given current and voltage are always the r.m.s. values.

Average Value

The average value of an alternating current is expressed by that steady current which transfers across any circuit the same charge as is transferred by that alternating current during the same time.

In the case of a symmetrical alternating current (i.e. one whose two half-cycles are exactly similar, whether sinusoidal or non-sinusoidal), the average value over a complete cycle is zero. Hence, in their case, the average value is obtained by adding or integrating the instantaneous values of current over one half-cycle only. But in the case of an unsymmetrical alternating current (like half-wave rectified current) the average value must always be taken over the whole cycle.

$$I_{av} = \frac{i_1 + i_2 + \dots + i_n}{n}$$

The standard equation of an alternating current is, $i = I_m \sin \theta$

$$I_{av} = \int_0^{\pi} \frac{id\theta}{(\pi - 0)} = \frac{I_m}{\pi} \int_0^{\pi} \sin \theta d\theta \quad (\text{putting value of } i)$$

$$= \frac{I_m}{\pi} \left| -\cos \theta \right|_0^{\pi} = \frac{I_m}{\pi} \left| +1 - (-1) \right| = \frac{2I_m}{\pi} = \frac{I_m}{\pi/2} = \frac{\text{twice the maximum current}}{\pi}$$

$$\therefore I_{av} = \frac{I_m}{\pi} \times \pi = 0.637 I_m \quad \therefore \text{average value of current} = 0.637 \times \text{maximum value}$$

Note. R.M.S. value is always greater than average value except in the case of a rectangular wave when both are equal.

Form Factor

It is defined as the ratio,

$$K_f = \text{r.m.s. value} / \text{average value}$$

$$= 0.707 I_m / 0.637 I_m$$

$$= 1.11 \text{ (for sinusoidal alternating currents only)}$$

In the case of sinusoidal alternating voltage also, $K_f = 0.707 V_m / 0.637 V_m = 1.11$

As is clear, the knowledge of form factor will enable the r.m.s. value to be found from the arithmetic mean value and vice-versa.

Crest or Peak or Amplitude Factor

It is defined as the ratio,

$$K_p = \text{maximum value} / \text{r.m.s. value}$$

$$= \frac{I_m}{I_m / \sqrt{2}} = \sqrt{2} = 1.414 \text{ (for sinusoidal a.c. only)}$$

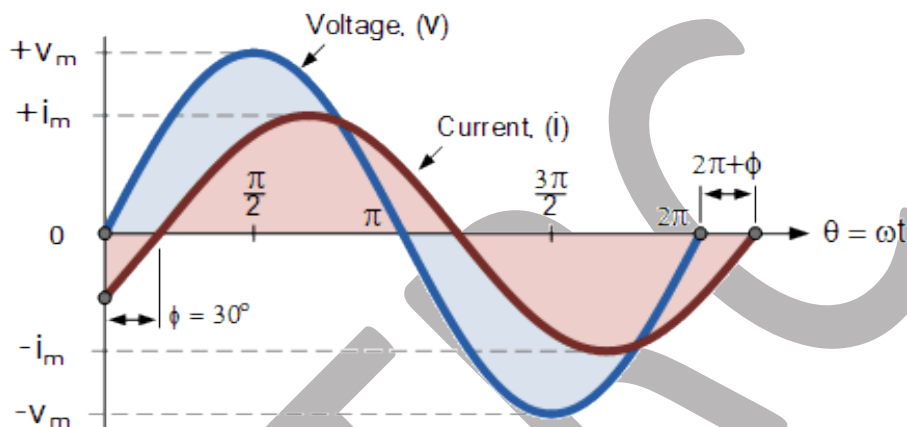
$$\text{For sinusoidal alternating voltage also, } K_a = \frac{E_m}{E_m / \sqrt{2}} = 1.414$$

LECTURE NO:-2

Phasor representation, real power, reactive power, apparent power, power factor

Phasor representation:

Phasor Diagrams are a graphical way of representing the magnitude and directional relationship between two or more alternating quantities



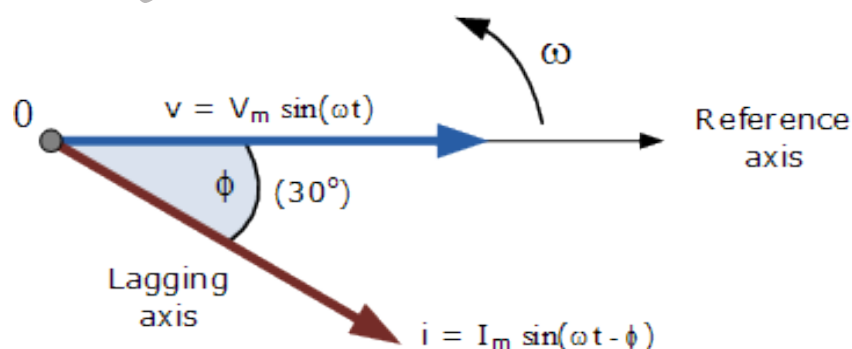
The generalised mathematical expression to define these two sinusoidal quantities will be written as:

$$v_{(t)} = V_m \sin(\omega t)$$

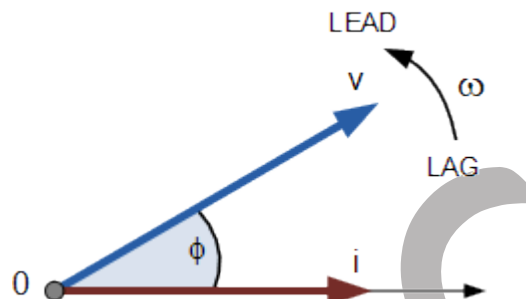
$$i_{(t)} = I_m \sin(\omega t - \phi)$$

The current, i is lagging the voltage, v by angle Φ and in our example above this is 30° . So the difference between the two phasors representing the two sinusoidal quantities is angle Φ and the resulting phasor diagram will be.

Phasor Diagram of a Sinusoidal Waveform



The phasor diagram is drawn corresponding to time zero ($t = 0$) on the horizontal axis. The lengths of the phasors are proportional to the values of the voltage, (V) and the current, (I) at the instant in time that the phasor diagram is drawn. The current phasor lags the voltage phasor by the angle, Φ , as the two phasors rotate in an anticlockwise direction as stated earlier, therefore the angle, Φ is also measured in the same anticlockwise direction.



If however, the waveforms are frozen at time, $t = 30^\circ$, the corresponding phasor diagram would look like the one shown on the right. Once again the current phasor lags behind the voltage phasor as the two waveforms are of the same frequency.

However, as the current waveform is now crossing the horizontal zero axis line at this instant in time we can use the current phasor as our new reference and correctly say that the voltage phasor is “leading” the current phasor by angle, Φ . Either way, one phasor is designated as the reference phasor and all the other phasors will be either leading or lagging with respect to this reference.

Active Power

The power which is actually consumed or utilised in an AC Circuit is called True power or Active power or Real power. It is measured in kilowatt (kW) or MW. It is the actual outcomes of the electrical system which runs the electric circuits or load.

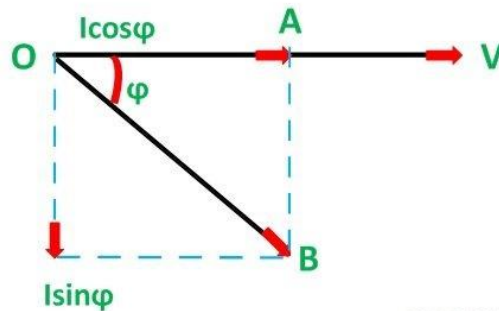
Reactive Power

The power which flows back and forth that means it moves in both the directions in the circuit or reacts upon itself, is called Reactive Power. The reactive power is measured in kilo volt-ampere reactive (kVAR) or MVAR.

Apparent Power

The product of root mean square (RMS) value of voltage and current is known as Apparent Power. This power is measured in kVA or MVA.

The phasor diagram for an inductive circuit is shown below:

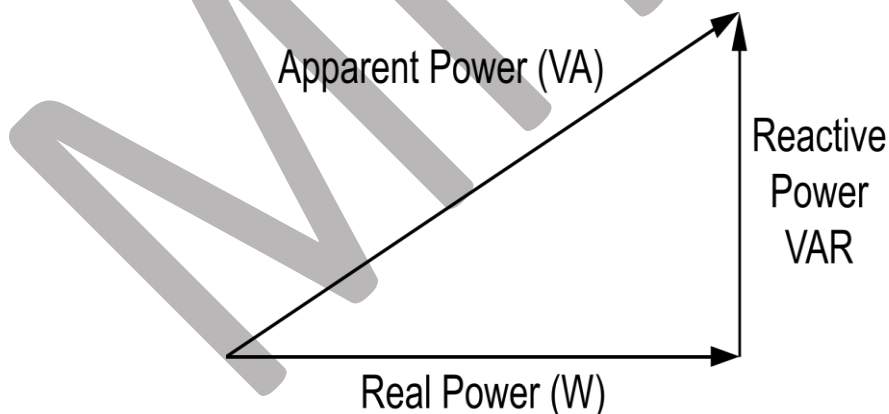


Taking voltage V as reference, the current I lags behind the voltage V by an angle ϕ . The current I is divided into two components:

- $I \cos \phi$ in phase with the voltage V
- $I \sin \phi$ which is 90 degrees out of phase with the voltage V

Therefore, the following expression shown below gives the active, reactive and apparent power respectively.

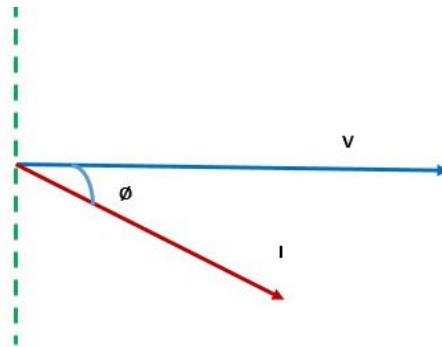
- Active power $P = V \times I \cos \phi = V I \cos \phi$
- Reactive power $Q = V \times I \sin \phi = V I \sin \phi$
- Apparent power $S = V \times I = VI$



Power Factor

Power factor is defined as the cosine of angle between the voltage phasor and current phasor in an AC circuit.

For an AC circuit, $0 \leq \text{pf} \leq 1$ whereas for DC circuit power factor is always 1



$$\text{pf} = \cos \phi$$

It can also be calculated from the power triangle. Since active power = $V \cos \phi$ and apparent power = VI , it can be calculated by calculating the ratio of Active Power and Apparent Power. The formula for pf in terms of active and apparent power is given as

$$\text{pf} = \text{Active Power} / \text{Apparent Power}$$

Power Factor for Resistance, Capacitor and Inductor:

The current flowing through the resistance is always in phase with the applied voltage. This means that the angle between them is 0° and hence

$$\text{pf} = \cos 0 = 1$$

This means that a resistance will only consume active power. It will not consume any reactive power.

Current through the inductor lags the applied voltage by 90° . Therefore the angle between them is 90° and hence

$$\text{pf} = \cos 90 = 0$$

Thus an inductor will not consume any active power. It only consumes reactive power.

Similarly, current through capacitor leads the voltage by 90° . Therefore

$$\text{pf} = \cos 90 = 0$$

A capacitor is thus generator of reactive power. It does not consume any active power.

LECTURE NO:-3

Analysis of single-phase AC circuits consisting of R, L, C

Pure Resistive Circuit:

$$V = V_m \sin \omega t \quad \text{--- ①}$$

$$i = \frac{V}{R}$$

$$i = \frac{V_m \sin \omega t}{R}$$

$$i = \left(\frac{V_m}{R} \right) \sin \omega t$$

$$i = I_m \sin \omega t \quad \text{--- ②}$$

$$\text{where } \left[I_m = \frac{V_m}{R} \right]$$

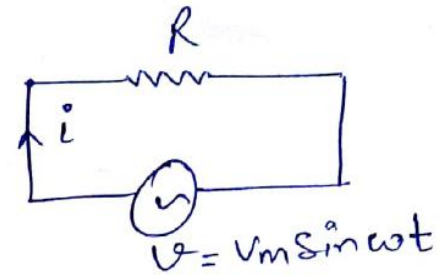


fig → ckt diagram.

From equation 1 & 2, it is evident that in a pure resistive circuit, the applied voltage and current are in phase with each other.



fig → Phasor Diagram

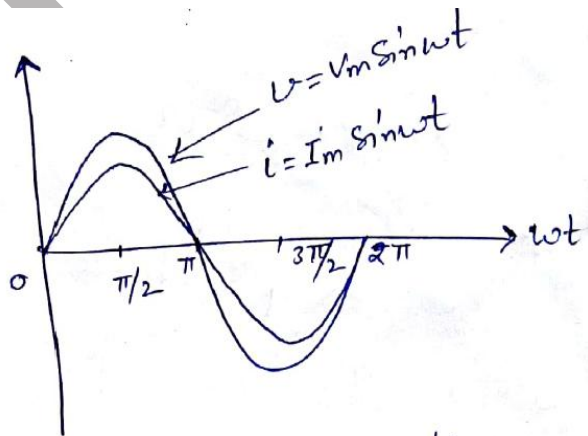
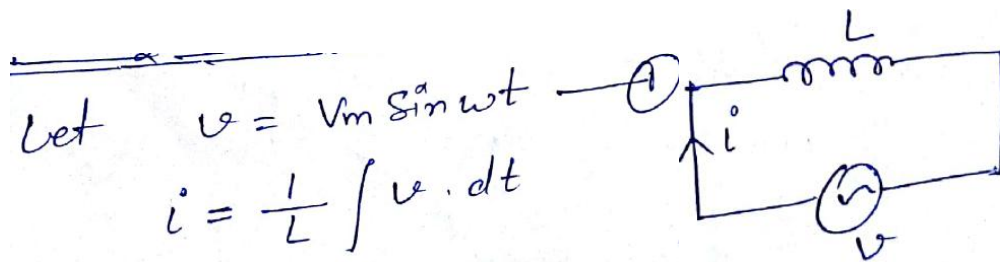


fig → wave diagram

Pure Inductive Circuit:

$$i = \frac{1}{L} \int V_m \sin \omega t \cdot dt$$

$$i = \frac{V_m}{L} \int \sin \omega t \cdot dt$$

$$= \frac{V_m}{L} \left[-\frac{\cos \omega t}{\omega} \right]$$

$$i = -\frac{V_m}{\omega L} \cos \omega t = \frac{V_m}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\boxed{i = I_m \sin \left(\omega t - \frac{\pi}{2} \right)} \quad \text{--- (2)}$$

where, $I_m = \frac{V_m}{\omega L}$

$$\boxed{X_L = \omega L} \quad \Omega$$

↓ Inductive Reactance

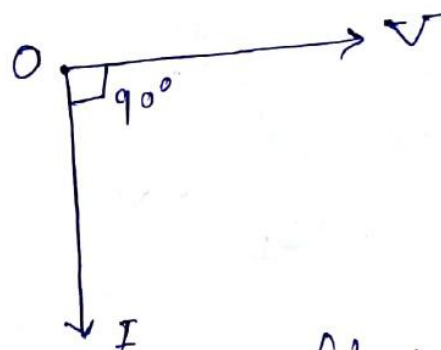


fig. 1 P. D.

Pure Capacitive Circuit:

Let $v = v_m \sin \omega t$ — (1)

Now, $i = C \frac{dv}{dt}$

$$i = C \frac{d}{dt} [v_m \sin \omega t]$$

$$i = \omega C v_m \cos \omega t$$

$$i = \frac{v_m}{1/\omega C} \sin(\omega t + 90^\circ) \quad \text{--- (2)}$$

$$i = I_m \sin(\omega t + \frac{\pi}{2})$$

where $I_m = \frac{v_m}{1/\omega C}$

$$X_C = \frac{1}{\omega C}$$

Capacitive Reactance

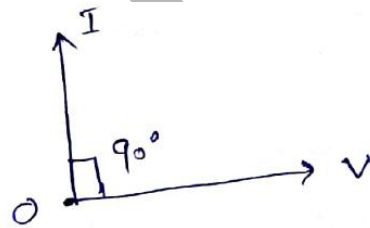
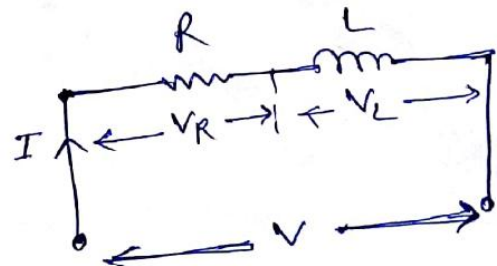


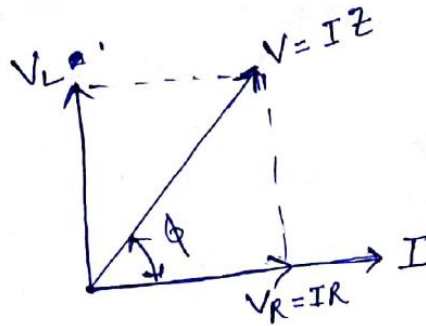
fig. → Phasor Diagram

* R-L Series CKT →

$$V_R = IR$$

$$V_L = I X_L$$





$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2} = IZ$$

$$Z = \sqrt{R^2 + X_L^2}$$

impedance

from the P.D., $\tan \phi = \frac{V_L}{V_R} = \frac{IX_L}{IR}$

$$\phi = \tan^{-1} \left(\frac{X_L}{R} \right)$$

if $V = V_m \sin \omega t$

then $i = I_m \sin(\omega t - \phi)$

where $I_m = \frac{V_m}{Z}$ and $\phi = \tan^{-1} \left(\frac{X_L}{R} \right)$

Power factor = $\cos \phi$

$$\therefore \cos \phi = \frac{R}{Z}$$

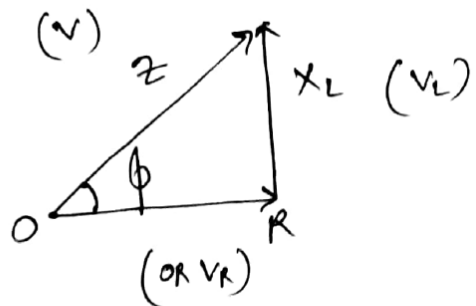


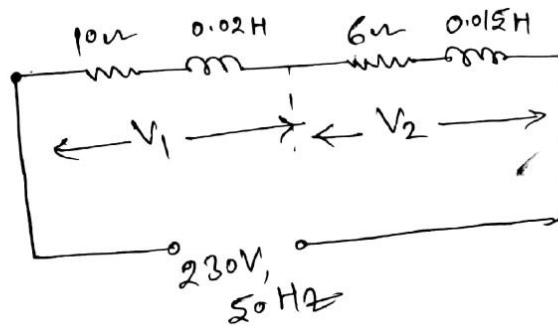
fig → Impedance Triangle

Example: A coil of resistance 10 ohm and inductance of 0.02 H is connected in series with another coil of resistance 6 ohm and inductance of 15 mH across 230 V, 50 Hz supply.

Calculate

- Impedance of the circuit
- Voltage drop across each coil
- Total power consumed by the circuit

Soln →



$$Z_1 = 10 + j2\pi \times 50 \times 0.02$$

$$= (10 + j6.283) \Omega = 11.81 \angle 32.14^\circ \Omega$$

$$Z_2 = (6 + j4.712) \Omega = 7.629 \angle 38.14^\circ \Omega$$

$$\therefore Z = Z_1 + Z_2 = (16 + j10.995) \Omega$$

$$= 19.414 \angle 34.5^\circ \Omega$$

$$I = \frac{V}{Z} = 11.847 \text{ A}$$

$$\left. \begin{aligned} V_1 &= I Z_1 = 139.9 \text{ V} \\ V_2 &= I Z_2 = 90.38 \text{ V} \end{aligned} \right\} \underline{\text{Ans}} \quad \textcircled{9}$$

LECTURE NO:-4

RL, RC and RLC combinations (series and parallel)

Example 1: A resistance of 20 ohm, inductance of 0.2 H and capacitance of 150 μF are connected in series and are fed by a 230 V, 50 Hz supply. Find X_L , X_C , Z , Y , p.f., active power and reactive power.

Solution.

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.2 = 62.8\Omega; X_C = 1/2\pi fC$$

$$= 10^{-6} 2\pi \times 50 \times 150 = 21.2\Omega; X = (X_L - X_C) = 41.6\Omega;$$

$$Z = \sqrt{R^2 + X^2} = \sqrt{20^2 + 41.6^2} = 46.2\Omega; I = V/Z = 230/46.2 = 4.98\text{ A}$$

Also, $Z = R + jX = 20 + j41.6 = 46.2 \angle 64.3^\circ \text{ ohm}$

$\therefore Y = 1/Z = 1/46.2 \angle 64.3^\circ = 0.0216 \angle -64.3^\circ \text{ siemens}$

p.f. = $\cos 64.3^\circ = 0.4336 \text{ (lag)}$

Active power = $VI \cos \phi = 230 \times 4.98 \times 0.4336 = 497\text{ W}$

Reactive power = $VI \sin \phi = 230 \times 4.98 \times \sin 64.3^\circ = 1031\text{ VAR}$

Example 2: A 120-V, 60-W lamp is to be operated on 220-V, 50-Hz supply mains. Calculate what value of (a) non-inductive resistance (b) pure inductance would be required in order that lamp is run on correct voltage. Which method is preferable and why?

Solution. Rated current of the bulb = $60/120 = 0.5\text{ A}$

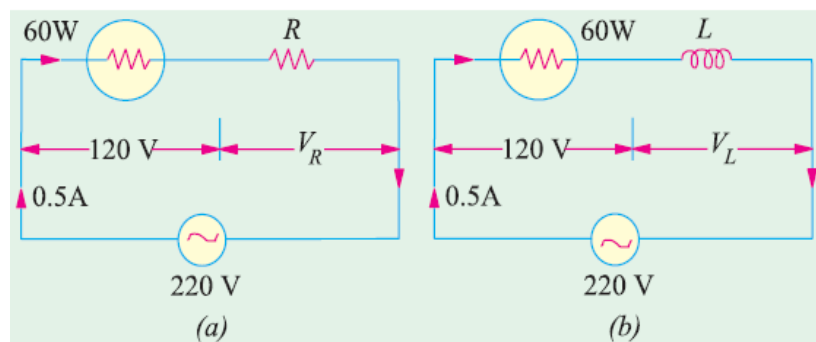


Fig. 13.15

(a) Resistor has been shown connected in series with the lamp in Fig. 13.15 (a).

P.D. across R is $V_R = 220 - 120 = 100\text{ V}$

It is in phase with the applied voltage, $\therefore R = 100/0.5 = 200\Omega$

(b) P.D. across bulb = 120 V
P.D. across L is

$$V_L = \sqrt{(220^2 - 120^2)} = 184.4\text{ V}$$

(Remember that V_L is in quadrature with V_R —the voltage across the bulb).

Now, $V_L = 0.5 \times X_L$ or $184.4 = 0.5 \times L \times 2\pi \times 50 \therefore L = 184.4/0.5 \times 3.14 = 1.17\text{ H}$

Method (b) is preferable to (a) because in method (b), there is no loss of power. Ohmic resistance of 200Ω itself dissipates large power (i.e. $100 \times 0.5 = 50\text{ W}$).

Example 3: A non-inductive resistor takes 8 A at 100 V. Calculate the inductance of a choke coil of negligible resistance to be connected in series in order that this load may be supplied from 220-V, 50-Hz mains. What will be the phase angle between the supply voltage and current

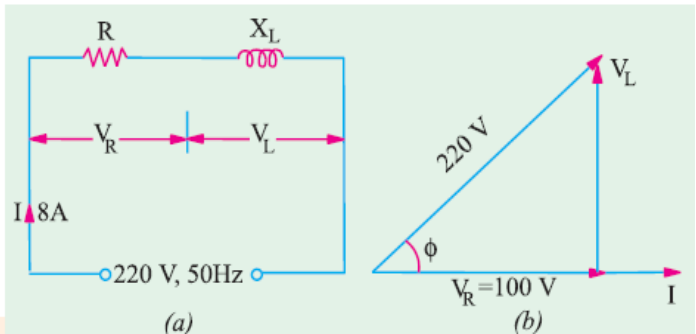
Solution. It is a case of pure resistance in series with pure inductance as shown in Fig. 13.16 (a).

Here $V_R = 100 \text{ V}$,

$$V_L = \sqrt{(220^2 - 100^2)} = 196 \text{ V}$$

Now, $V_L = I \cdot X_L$

or $196 = 8 \times 2\pi \times 50 \times L = \mathbf{0.078 \text{ H}}$



Example 4: Two coils A and B are connected in series across a 240-V, 50-Hz supply. The resistance of A is 5Ω and the inductance of B is 0.015 H . If the input from the supply is 3 kW and 2 kVAR , find the inductance of A and the resistance of B. Calculate the voltage across each coil.

Solution. The kVA triangle is shown in Fig. 13.18 (b) and the circuit in Fig. 13.18(a). The circuit kVA is given by, $\text{kVA} = \sqrt{(3^2 + 2^2)} = 3.606$ or $VA = 3,606$ voltamperes

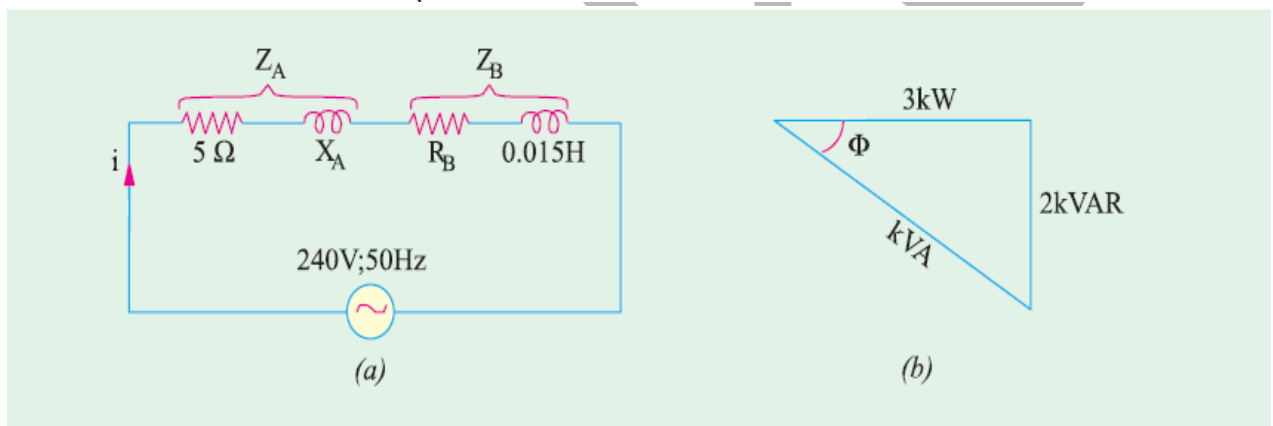


Fig. 13.18

Circuit current $= 3,606/240 = 15.03 \text{ A}$ $\therefore 15.03^2 (R_A + R_B) = 3,000$

$\therefore R_A + R_B = 3,000/15.03^2 = 13.3 \Omega$ $\therefore R_B = 13.3 - 5 = \mathbf{8.3 \Omega}$

Now, impedance of the whole circuit is given by $Z = 240/15.03 = 15.97 \Omega$

$\therefore X_A + X_B = \sqrt{(Z^2 - (R_A + R_B)^2)} = \sqrt{15.97^2 - 13.3^2} = 8.84 \Omega$

Now $X_B = 2\pi \times 50 \times 0.015 = 4.713 \Omega$ $\therefore X_A = 8.843 - 4.713 = 4.13 \Omega$

or $2\pi \times 50 \times L_A = 4.13$ $\therefore L_A = \mathbf{0.0132 \text{ H (approx)}}$

Now $Z_A = \sqrt{R_A^2 + X_A^2} = \sqrt{5^2 + 4.13^2} = 6.585 \Omega$

P.D. across coil A $= I \cdot Z_A = 15.03 \times 6.485 = \mathbf{97.5 \text{ V}}$; $Z_B = \sqrt{8.3^2 + 4.713^2} = 9.545 \Omega$

$\therefore$ p.d. across coil B $= I \cdot Z_B = 15.03 \times 9.545 = \mathbf{143.5 \text{ V}}$

Example 5: Resistor ($= R$), choke-coil (r, L), and a capacitor of $25.2 \mu\text{F}$ are connected in series. When supplied from an A.C. source, it takes 0.4 A . If the voltage across the resistor is 20 V , voltage across the resistor and choke is 45 volts , voltage across the choke is 35 volts , and voltage across the capacitor is 50 V .

Find : (a) The values of r, L

(b) Applied voltage and its frequency,

(c) P.F. of the total circuit and active power consumed. Draw the phasor diagram.

Solution.

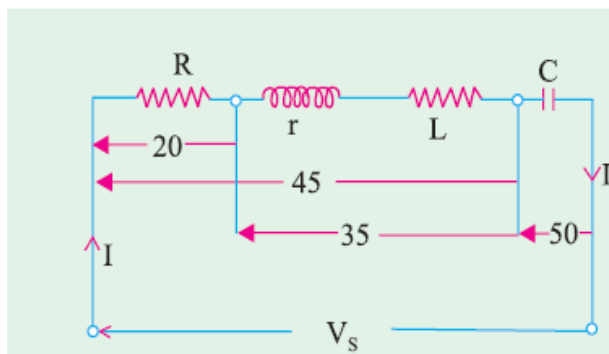


Fig. 13.23 (a)

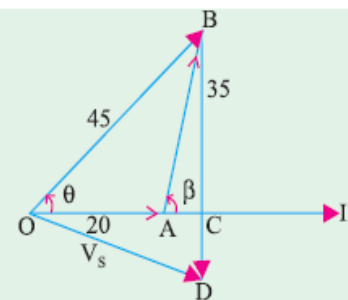


Fig. 13.23 (b)

(b) Since the current I is 0.4 amp , and voltage drop across R is 20 V ,

$$R = 20/0.4 = 50 \text{ ohms}$$

Similarly, Impedance of the coil, $Z_1 = 35/0.4 = 87.5 \text{ ohms}$

Capacitive reactance $X_c = 50/0.4 = 125 \text{ ohms}$

With a capacitor of $25.5 \mu\text{F}$, and supply angular frequency of ω radians/sec

$$\frac{1}{\omega} = X_C \cdot C = 125 \times 25.5 \times 10^{-6}, \text{ which gives } \omega = 314 \text{ rad./sec.}$$

The corresponding source frequency, $f = 50 \text{ Hz}$

(c) The phasor diagram is drawn in Fig. 13.23 (b), taking I as the reference.

Solving triangle OAB ,

$$\cos \phi = \frac{45^2 + 20^2 - 35^2}{2 \times 45 \times 20} = 0.667, \text{ and hence } \phi = 48.2^\circ$$

$$\text{Similarly, } \cos (180^\circ - \beta) = \frac{400 + 1225 - 2025}{2 \times 20 \times 35} = 106.6^\circ$$

This gives $\beta = 73.4^\circ$. From the phasor diagram is Fig. 13.23 (b),

$$OC = OA + AC = 20 + 35 \cos \beta = 30$$

$$BC = 35 \sin \beta = 33.54. \text{ The capacitive-reactance drop is } BD.$$

Since $BD = 50$, $CD = 16.46$ volts.

$$V_s = \sqrt{OC^2 + CD^2} = 34.22 \text{ volts}$$

$$\angle COD = \phi = \cos^{-1} (OC/OD) = 28.75^\circ$$

The power-factor of the total circuit = $\cos \phi = 0.877$, Leading,

Since I Leads V_s in Fig. 13.23 (b).

(a) For the coil, ACB part of the phasor diagram is to be observed.

$$r = AC/I = 10/0.4 = 25 \text{ ohms}$$

$$X_L = BC/I = 33.54/0.4 = 83.85 \text{ ohms}$$

Hence, coil-inductance, $L = 83.85/314 = 267$ milli-henry.

$$P = \text{Active Power Consumed} = V_s I \cos \phi = 12 \text{ watts}$$

$$\text{or } P = (0.4)^2 \times (R + r) = 12 \text{ watts}$$

LECTURE NO:-5

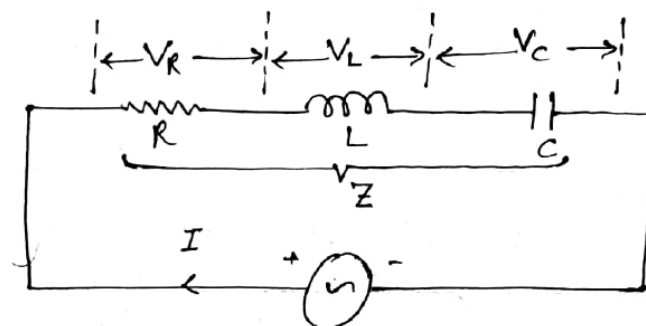
Resonance- Series Resonance

Resonance:

At resonance, the inductive reactance is equal to capacitive reactance and hence the voltage across inductor and capacitor cancel each other. The total impedance of circuit is resistance only. So, the circuit behaves like a pure resistive circuit and we know that in pure resistive circuit, voltage and the circuit current are in same phase i.e. V_r , V and I are in same phase direction. Therefore, the phase angle between voltage and current is zero and the power factor is unity.

Resonance in Series RLC Circuit

* Series Resonance $\rightarrow$



$$I = \frac{V}{Z}$$

$Z \rightarrow$ Equivalent impedance of the ckt

$$Z = R + j\omega L + \frac{1}{j\omega C}$$

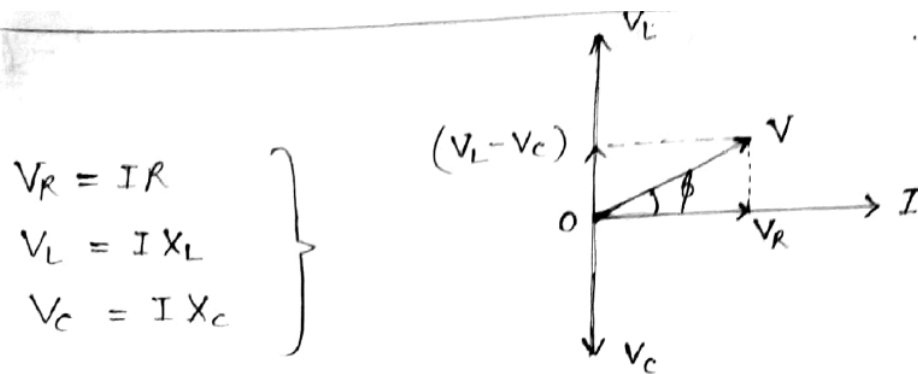
$$= R + j\omega L - \frac{j}{\omega C}$$

OR $Z = R + j(X_L - X_C)$

$$\left[\begin{array}{l} X_L = \omega L \\ X_C = \frac{1}{\omega C} \end{array} \right]$$

Thus, $I = \frac{V}{R + j(X_L - X_C)}$

$$I = \frac{V}{R + jX} \quad A$$



Resonance condition is obtained when $X_L = X_C$ in the series ckt.

When $X_L = X_C$, $Z = R + j(0)$
 $Z = R$

$$I_0 = \frac{V}{R} \quad \left[I_0 \rightarrow \text{current at resonance} \right]$$

Pf, $\cos \phi = \frac{R}{Z} = \frac{R}{R} = 1$

Let ω_0 or f_0 be the frequency at which

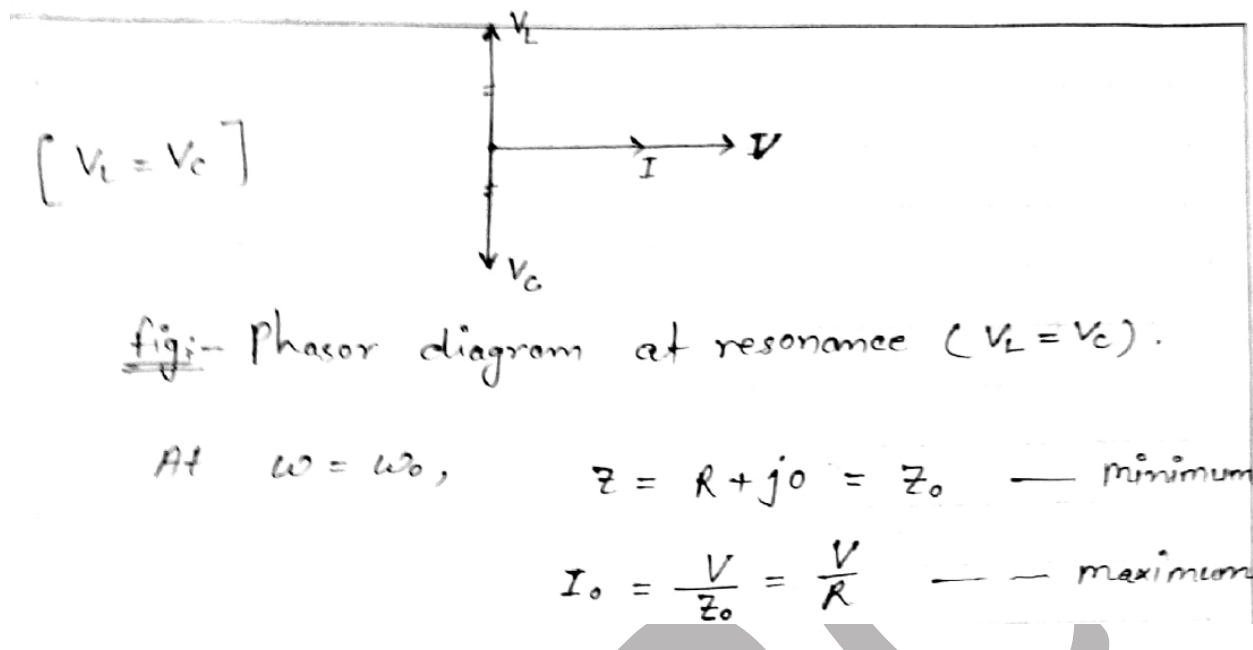
$$X_L = X_C \quad \left[\omega_0 \rightarrow \text{Resonant frequency} \right]$$

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0^2 = \frac{1}{LC} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}}$$



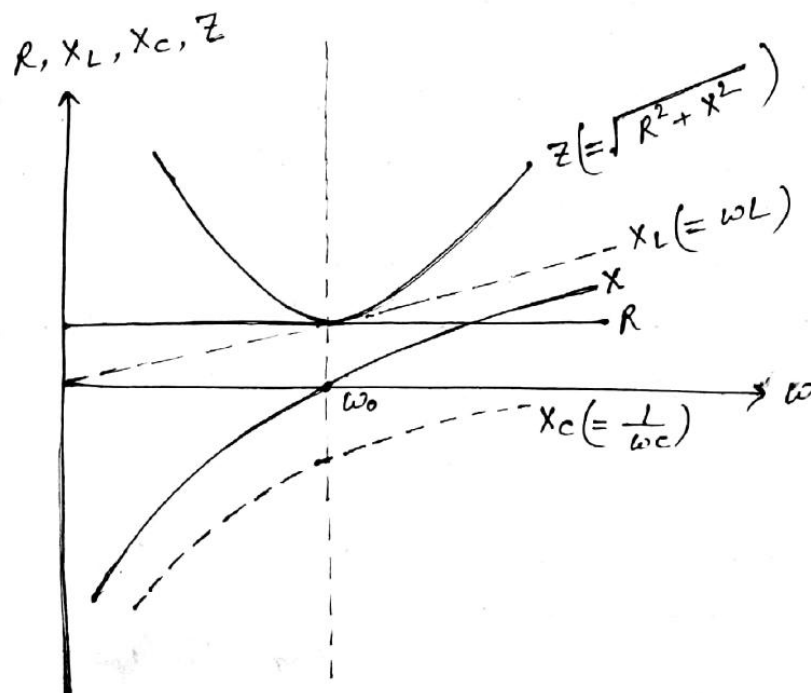
Summary

For a series RLC circuit at certain frequency called resonant frequency, the following points must be remembered. So at resonance:

1. Inductive reactance X_L is equal to capacitive reactance X_C .
2. Total impedance of circuit becomes minimum which is equal to R , i.e. $Z = R$.
3. Circuit current becomes maximum as impedance reduces, $I = V / R$.
4. Voltage across inductor and capacitor cancels each other, so voltage across resistor $V_r = V$, supply voltage.
5. Since net reactance is zero, circuit becomes purely resistive circuit and hence the voltage and the current are in same phase, so the phase angle between them is zero.
6. Power factor is unity.
7. Frequency at which resonance in series RLC circuit occurs is given by

$$\frac{1}{2\pi\sqrt{LC}} = f_r \text{ Hz}$$

Variation of circuit parameters with frequency:



Example: An R-L-C series circuit consists of a resistance of $1000\ \Omega$, an inductance of 100 mH and a capacitance of $10\ \mu\text{F}$. If a voltage of 100 V is applied across the combination. Find (i) the resonance frequency (ii) Q-factor of the circuit and (iii) the half-power points.

Solution. (i)
$$f_0 = \frac{1}{2\pi \sqrt{L \times C}} = \frac{10^6}{2\pi} = 159\text{ kHz}$$

(ii)
$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{1000} \times \sqrt{\frac{10^{-1}}{10^{-11}}} = 100$$

(iii)
$$f_1 = f_0 - \frac{R}{4\pi L} = 159 \times 10^3 - \frac{1000}{4\pi \times 10^{-1}} = 158.2\text{ kHz}$$

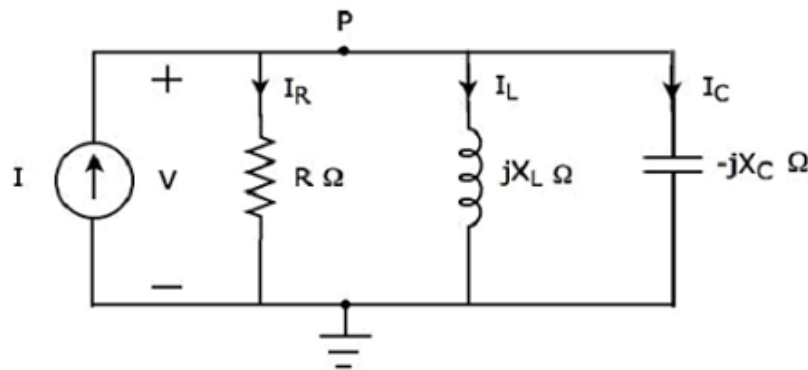
$$f_2 = f_0 + \frac{R}{4\pi L} = 159 \times 10^3 + \frac{1000}{4\pi \times 10^{-1}} = 159.8\text{ kHz}$$

LECTURE NO:-6

Parallel Resonance

Resonance in Parallel RLC Circuit:

If the resonance occurs in parallel RLC circuit, then it is called as Parallel Resonance. Consider the following parallel RLC circuit, which is represented in phasor domain.



We know from the previous series resonance tutorial that resonance takes place when $V_L = -V_C$ and this situation occurs when the two reactances are equal, $X_L = X_C$. The admittance of a parallel circuit is given as:

$$Y = G + B_L + B_C$$

$$Y = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$$

or

$$Y = \frac{1}{R} + \frac{1}{2\pi f L} + 2\pi f C$$

Resonance occurs when $X_L = X_C$ and the imaginary parts of Y become zero. Then:

$$X_L = X_C \Rightarrow 2\pi f L = \frac{1}{2\pi f C}$$

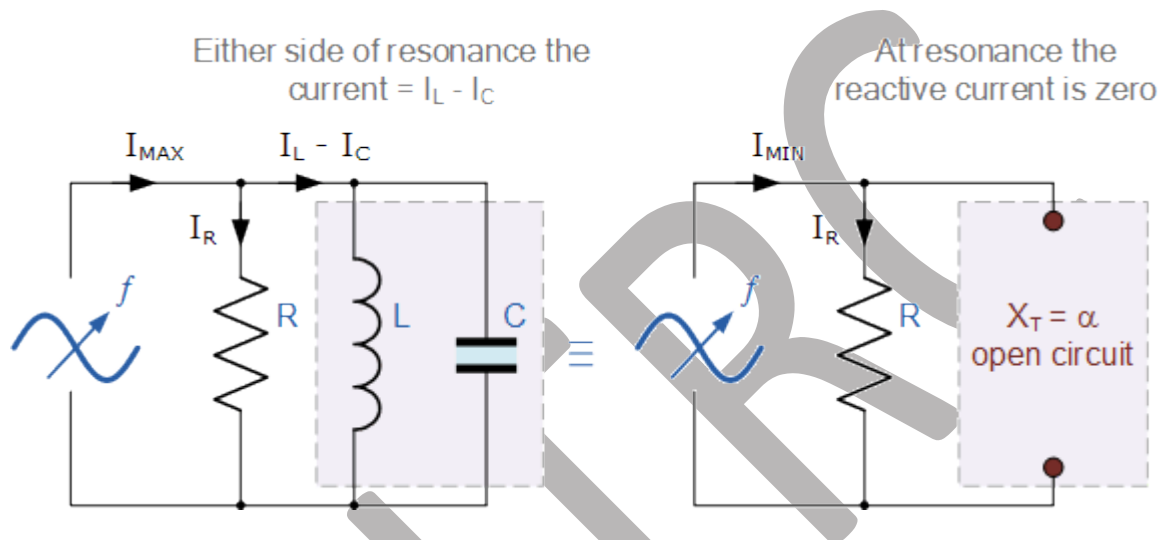
$$f^2 = \frac{1}{2\pi L \times 2\pi C} = \frac{1}{4\pi^2 LC}$$

$$f = \sqrt{\frac{1}{4\pi^2 LC}}$$

$$\therefore f_r = \frac{1}{2\pi \sqrt{LC}} \text{ (Hz)} \quad \text{or} \quad \omega_r = \frac{1}{\sqrt{LC}} \text{ (rads)}$$

Notice that at resonance the parallel circuit produces the same equation as for the series resonance circuit. Therefore, it makes no difference if the inductor or capacitor are connected in parallel or series.

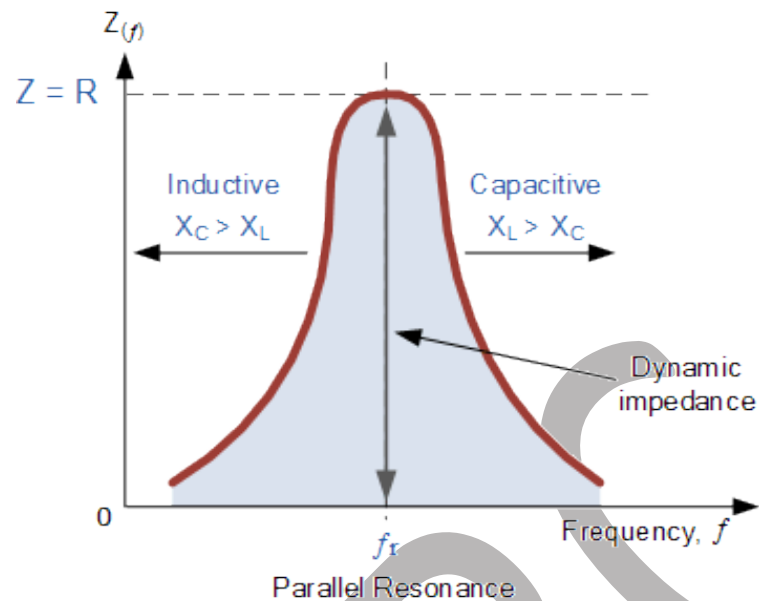
Also at resonance the parallel LC tank circuit acts like an open circuit with the circuit current being determined by the resistor, R only. So the total impedance of a parallel resonance circuit at resonance becomes just the value of the resistance in the circuit and $Z = R$ as shown.



Thus at resonance, the impedance of the parallel circuit is at its maximum value and equal to the resistance of the circuit creating a circuit condition of high resistance and low current. Also at resonance, as the impedance of the circuit is now that of resistance only, the total circuit current, I will be “in-phase” with the supply voltage, V_S .

We can change the circuit’s frequency response by changing the value of this resistance. Changing the value of R affects the amount of current that flows through the circuit at resonance, if both L and C remain constant. Then the impedance of the circuit at resonance $Z = R_{MAX}$ is called the “dynamic impedance” of the circuit.

Impedance in a Parallel Resonance Circuit



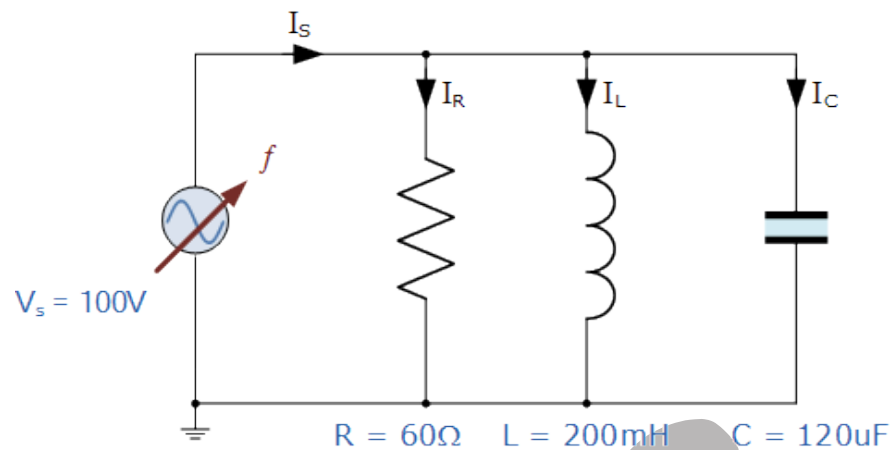
Note that if the parallel circuits impedance is at its maximum at resonance then consequently, the circuits admittance must be at its minimum and one of the characteristics of a parallel resonance circuit is that admittance is very low limiting the circuits current. Unlike the series resonance circuit, the resistor in a parallel resonance circuit has a damping effect on the circuits bandwidth making the circuit less selective.

Also, since the circuit current is constant for any value of impedance, Z , the voltage across a parallel resonance circuit will have the same shape as the total impedance and for a parallel circuit the voltage waveform is generally taken from across the capacitor.

We now know that at the resonant frequency, f_r , the admittance of the circuit is at its minimum and is equal to the conductance, G given by $1/R$ because in a parallel resonance circuit the imaginary part of admittance, i.e. the susceptance, B is zero because $B_L = B_C$ as shown.

Example:

A parallel resonance network consisting of a resistor of 60Ω , a capacitor of $120\mu\text{F}$ and an inductor of 200mH is connected across a sinusoidal supply voltage which has a constant output of 100 volts at all frequencies. Calculate, the resonant frequency, the quality factor and the bandwidth of the circuit, the circuit current at resonance and current magnification.



1. Resonant Frequency, f_r

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.2 \cdot 120 \cdot 10^{-6}}} = 32.5\text{Hz}$$

2. Inductive Reactance at Resonance, X_L

$$X_L = 2\pi fL = 2\pi \cdot 32.5 \cdot 0.2 = 40.8\Omega$$

3. Quality factor, Q

$$Q = \frac{R}{X_L} = \frac{R}{2\pi fL} = \frac{60}{40.8} = 1.47$$

4. Bandwidth, BW

$$BW = \frac{f_r}{Q} = \frac{32.5}{1.47} = 22\text{Hz}$$

5. The upper and lower -3dB frequency points, f_H and f_L

$$f_L = f_r - \frac{1}{2}BW = 32.5 - \frac{1}{2}(22) = 21.5\text{Hz}$$

$$f_H = f_r + \frac{1}{2}BW = 32.5 + \frac{1}{2}(22) = 43.5\text{Hz}$$

6. Circuit Current at Resonance, I_T

At resonance the dynamic impedance of the circuit is equal to R

$$I_T = I_R = \frac{V}{R} = \frac{100}{60} = 1.67A$$

7. Current Magnification, I_{mag}

$$I_{MAG} = Q \times I_T = 1.47 \times 1.67 = 2.45A$$

Note that the current drawn from the supply at resonance (the resistive current) is only 1.67 amps, while the current flowing around the LC tank circuit is larger at 2.45 amps. We can check this value by calculating the current flowing through the inductor (or capacitor) at resonance.

$$I_L = \frac{V}{X_L} = \frac{V}{2\pi fL} = \frac{100}{2\pi \cdot 32.5 \cdot 0.2} = 2.45A$$

LECTURE NO:-7

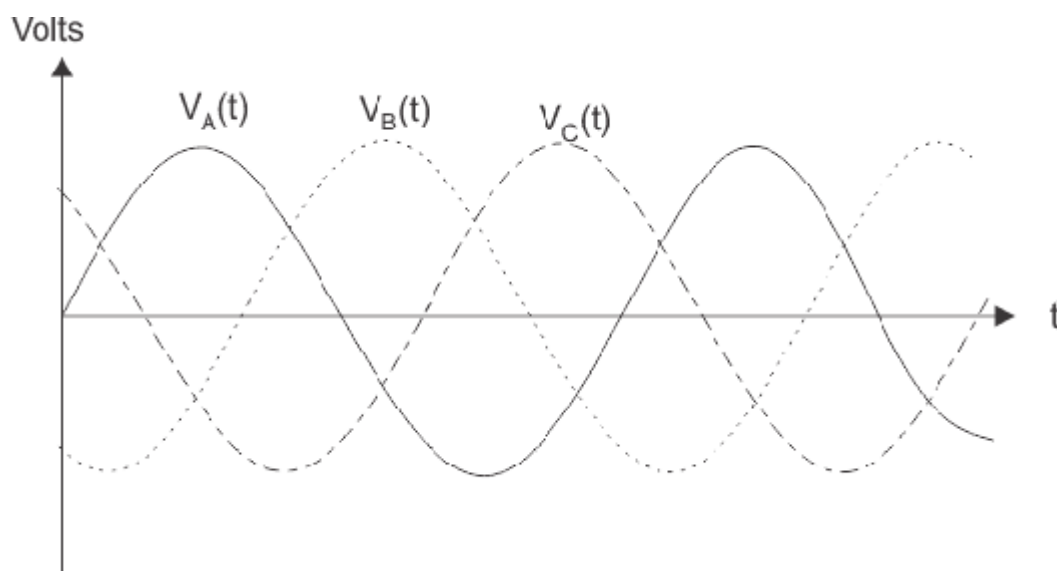
Three Phase Balanced Circuits

Introduction:

There are two types of system available in electric circuit, single phase and three phasesystem. In single phase circuit, there will be only one phase, i.e the current will flow through only one wire and there will be one return path called neutral line to complete the circuit. So in single phase minimum amount of power can be transported. Here the generating station and load station will also be single phase.

In 1882, new invention has been done on polyphase system, that more than one phase can be used for generating, transmitting and for load system. Three phase circuit is the polyphase system where three phases are send together from the generator to the load. Each phase are having a phase difference of 120° , i.e 120° angle electrically. So from the total of 360° , three phases are equally divided into 120° each. The power in three phasesystem is continuous as all the three phases are involved in generating the total power. The sinusoidal waves for 3 phase system is shown below-

The three phases can be used as single phase each. So if the load is single phase, then one phase can be taken from the three phase circuit and the neutral can be used as ground to complete the circuit.

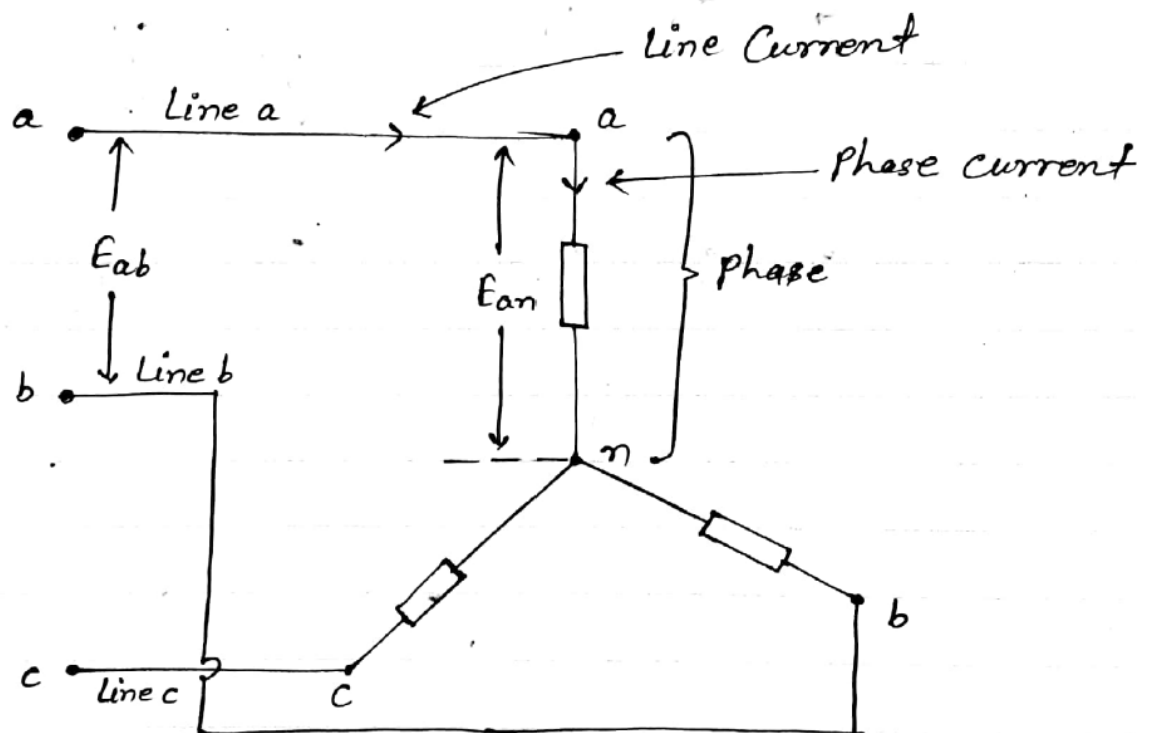
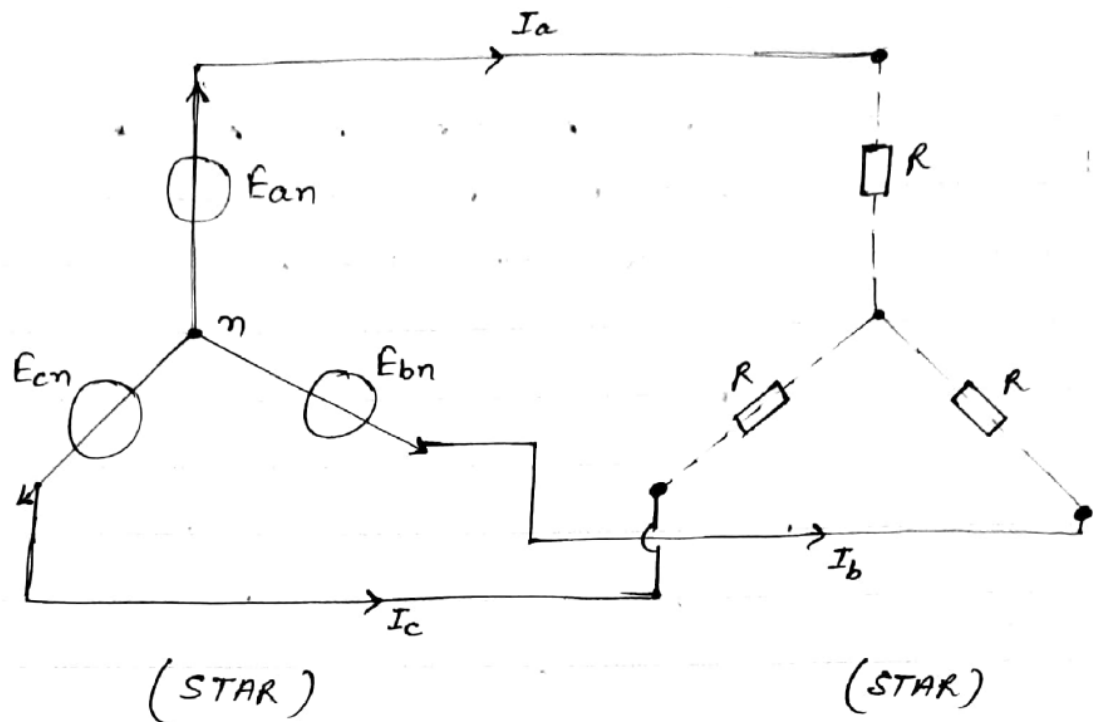


Advantages of Three Phase System over Single Phase System:

Presently 3- ϕ AC system is very popular and being used worldwide for power generation, power transmission, distribution and for electric motors.

Three phase system has the following advantages as compare to single phase system:

1. For electric power transmission and distribution of same amount of power, the requirement of conductor material is less in 3- ϕ system as compare to 1- ϕ system. Hence, the 3- ϕ transmission and distribution system is economical as compare 1- ϕ system.
2. 3-phase induction motor is self-started as the magnetic flux produced by 3-phase supply is rotating in nature with a constant magnitude. Whereas 1- ϕ induction motor is not self-started as the magnetic flux produced by 1- ϕ supply is pulsating in nature. Hence, we have to make some arrangement to make the 1- ϕ induction motor self-started which further increases the cost of 1- ϕ induction motor.
3. 3-phase motor is having better power factor
4. Power to weight ratio of 3-phase transformer is high as compare to 1- ϕ transformer. Means for the same amount of Electric Power, the size of 3-phase transformer is small as compared to 1- ϕ transformer. Hence, the overall cost of a transformer gets reduced. Moreover, due to reduction in weight, transportation and installation of transformer become convenient and less space is required to accommodate the transformer.
5. If a fault occurs in any winding of a 3-phase transformer, the rest of two winding can be used in the open delta to serve the 3-phase load. The same is not possible in 1- ϕ transformer. This ability of 3-phase transformer further increases the reliability of 3-phase transformer.
6. A 3-phase system can be used to feed a 1- ϕ load, whereas vice-versa is not possible.



E_{ab}
 E_{bc}
 E_{ca}

}

line voltages.

E_{an}
 E_{bn}
 E_{cn}

}

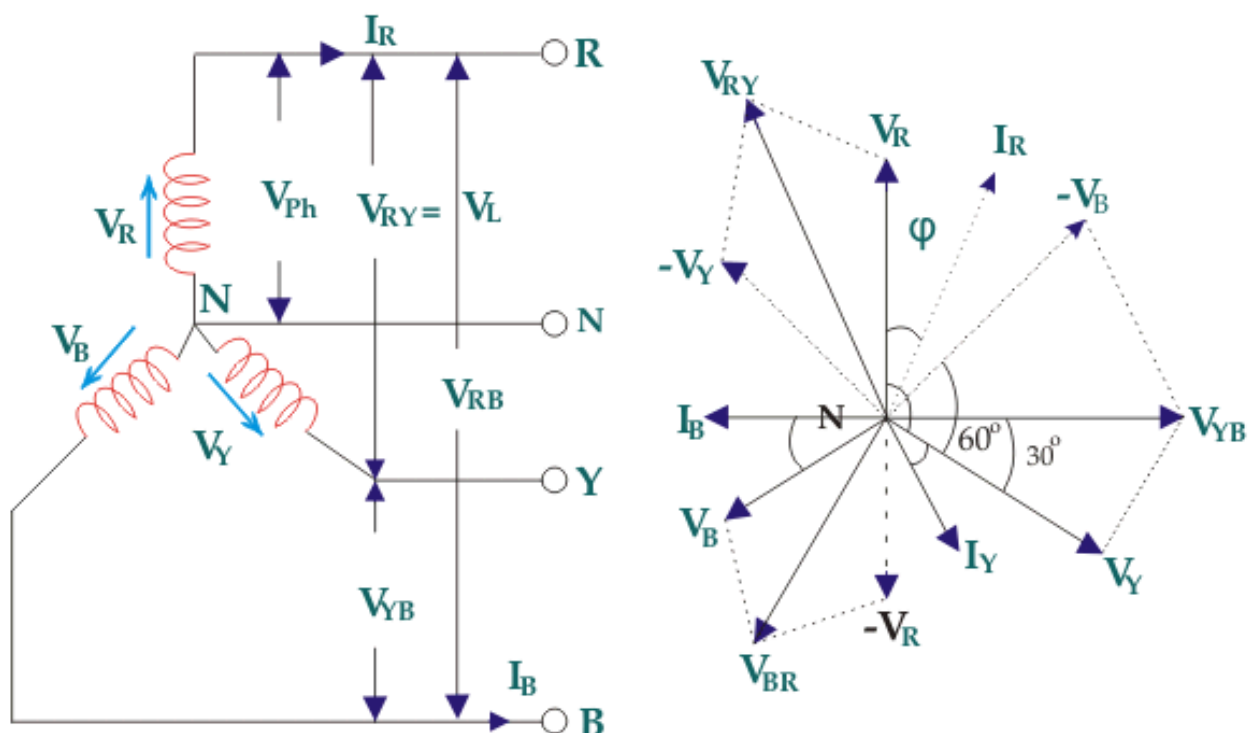
phase voltages.

LECTURE NO:-8

Voltage and Current Relations in Star and Delta Connections

Relationship of Line and Phase Voltages and Currents in a Star Connected System:

To derive the relations between line and phase currents and voltages of a star connected system, we have first to draw a balanced star connected system.



Suppose due to load impedance the current lags the applied voltage in each phase of the system by an angle ϕ . As we have considered that the system is perfectly balanced, the magnitude of current and voltage of each phase is the same. Let us say, the magnitude of the voltage across the red phase i.e. magnitude of the voltage between neutral point (N) and red phase terminal (R) is V_R .

Similarly, the magnitude of the voltage across yellow phase is V_Y and the magnitude of the voltage across blue phase is V_B .

In the balanced star system, magnitude of phase voltage in each phase is V_{ph} .

$$\therefore V_R = V_Y = V_B = V_{ph}$$

We know in the star connection, line current is same as phase current. The magnitude of this current is same in all three phases and say it is I_L .

$\therefore I_R = I_Y = I_B = I_L$, Where, I_R is line current of R phase, I_Y is line current of Y phase and I_B is line current of B phase. Again, phase current, I_{ph} of each phase is same as line current I_L in star connected system.

$$\therefore I_R = I_Y = I_B = I_L = I_{ph}.$$

Now, let us say, the voltage across R and Y terminal of the star connected circuit is V_{RY} . The voltage across Y and B terminal of the star connected circuit is V_{YB} .

From the diagram,
it is found that

$$V_{RY} = V_R + (-V_Y)$$

$$\text{Similarly, } V_{YB} = V_Y + (-V_B)$$

$$\text{And, } V_{BR} = V_B + (-V_R)$$

Now, as angle between V_R and V_Y is 120° (electrical), the angle between V_R and $-V_Y$ is $180^\circ - 120^\circ = 60^\circ$ (electrical).

$$\begin{aligned} V_L = |V_{RY}| &= \sqrt{V_R^2 + V_Y^2 + 2V_R V_Y \cos 60^\circ} \\ &= \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph} V_{ph} \times \frac{1}{2}} \\ &= \sqrt{3} V_{ph} \\ \therefore V_L &= \sqrt{3} V_{ph} \end{aligned}$$

Thus, for the star-connected system line voltage = $\sqrt{3} \times$ phase voltage.

Line current = Phase current

As, the angle between voltage and current per phase is ϕ , the electric power per phase is

$$V_{ph} I_{ph} \cos \phi = \frac{V_L}{\sqrt{3}} I_L \cos \phi$$

So the total power of three phase system is

$$3 \times \frac{V_L}{\sqrt{3}} I_L \cos \phi = \sqrt{3} V_L I_L \cos \phi$$

Relation between Phase Voltage and Line Voltage in Delta Connection

To understand the relationship between the phase voltage and line voltage in the Delta consider the figure A shown below.

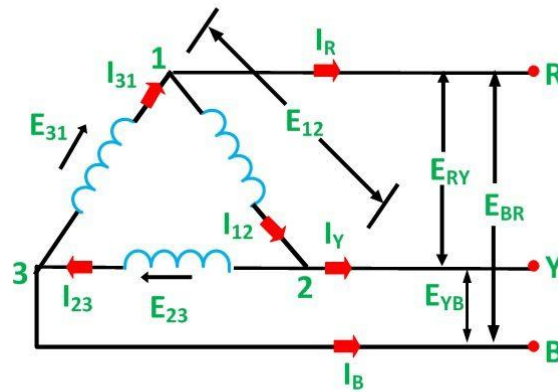


Figure A

Circuit Globe

It is clear from the figure that the voltage across terminals 1 and 2 is the same as across the terminals R and Y. Therefore,

$$E_{12} = E_{RY}$$

Similarly,

$$E_{23} = E_{YB} \quad \text{and} \quad E_{31} = E_{BR}$$

Where, the phase voltages are

$$E_{12} = E_{23} = E_{31} = E_{ph}$$

The line voltages are

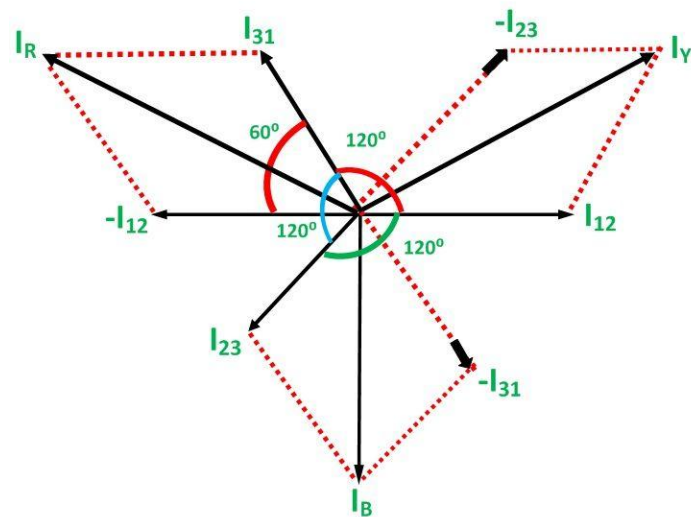
$$E_{RY} = E_{YB} = E_{BR} = E_L$$

Hence, in Delta Connection Line Voltage is equal to Phase Voltage.

Relation between Phase Current and Line Current in Delta Connection

As in the balanced system the three phase current I_{12} , I_{23} and I_{31} are equal in magnitude but are displaced from one another by 120 degrees electrical.

The **phasor diagram** is shown below.



Hence,

$$I_{12} = I_{23} = I_{31} = I_{ph}$$

If we look at figure A, it is seen that the current is divided at every junction 1, 2 and 3. Applying Kirchhoff's Law at junction 1

The Incoming currents are equal to outgoing currents.

$$\overline{I_{31}} = \overline{I_R} + \overline{I_{12}}$$

And their vector difference will be given as

$$\overline{I_R} = \overline{I_{31}} - \overline{I_{12}}$$

The vector I_{12} is reversed and is added in the vector I_{31} to get the vector sum of I_{31} and $-I_{12}$ as shown above in the phasor diagram. Therefore,

$$I_R = \sqrt{I_{31}^2 + I_{12}^2 + 2I_{31}I_{12} \cos 60^\circ} \text{ or}$$

$$I_L = \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}I_{ph} \times 0.5}$$

As we know, $I_R = I_L$, therefore,

$$I_L = \sqrt{3I_{ph}^2} = \sqrt{3}I_{ph}$$

Similarly,

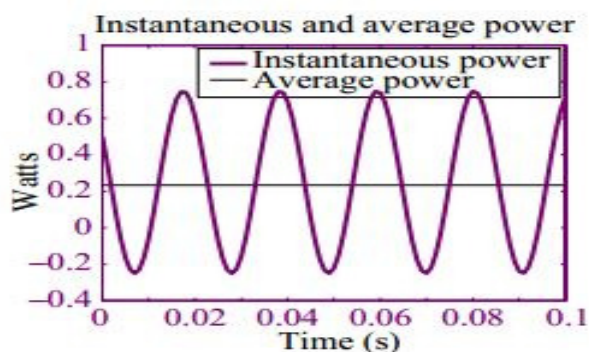
$$\overline{I}_Y = \overline{I}_{12} - \overline{I}_{23} \text{ or } I_L = \sqrt{3}I_{ph} \text{ and}$$

$$\overline{I}_B = \overline{I}_{23} - \overline{I}_{31} \text{ or } I_L = \sqrt{3}I_{ph}$$

Hence, in Delta connection line current is root three times of phase current.

$$\text{Line Current} = \sqrt{3} \times \text{Phase Current}$$

LECTURE-9 POWER FACTOR & COMPLEX POWER



In phasor notation, the current and voltage are given by

$$\mathbf{V}(j\omega) = V e^{j\theta}$$

$$\mathbf{I}(j\omega) = I e^{-j\theta}$$

impedance of the circuit element defined by the phasor voltage and current to be

$$Z = \frac{V}{I} e^{-j(\theta)} = |Z| e^{j\theta_Z}$$

The expression for the average power using phasor notation

$$P_{av} = \frac{1}{2} \frac{V^2}{|Z|} \cos \theta = \frac{1}{2} I^2 |Z| \cos \theta$$

Power Factor

The phase angle of the load impedance plays a very important role in the absorption of power by load impedance. The average power dissipated by an AC load is dependent on the cosine of the angle of the impedance. To recognize the importance of this factor in AC power computations, the term $\cos(\theta)$ is referred to as the power factor (pf). Note that the power factor is equal to 0 for a purely

inductive or capacitive load and equal to 1 for a purely resistive load; in every other case, $0 < \text{pf} < 1$. If the load has an inductive reactance, then θ is positive and the current lags (or follows) the voltage. Thus, when θ and Q are positive, the corresponding power factor is termed lagging. Conversely, a capacitive load will have a negative Q , and hence a negative θ . This corresponds to a leading power factor, meaning that the load current leads the load voltage. A power factor close to unity signifies an efficient transfer of energy from the AC source to the load, while a small power factor corresponds to inefficient use of energy. Two equivalent expressions for the power factor are given in the following:

$$\text{pf} = \cos(\theta) = \frac{P_{\text{av}}}{\tilde{V}\tilde{I}} \quad \text{Power factor}$$

where $\tilde{V}$ and $\tilde{I}$ are the rms values of the load voltage and current.

Complex Power

The expression for the instantaneous power may be further expanded to provide further insight into AC power. Using trigonometric identities, we obtain the

$$\begin{aligned} p(t) &= \frac{\tilde{V}^2}{|Z|} [\cos \theta + \cos \theta \cos(2\omega t) + \sin \theta \sin(2\omega t)] \\ &= \tilde{I}^2 |Z| [\cos \theta + \cos \theta \cos(2\omega t) + \sin \theta \sin(2\omega t)] \\ &= \tilde{I}^2 |Z| \cos \theta (1 + \cos(2\omega t)) + \tilde{I}^2 |Z| \sin \theta \sin(2\omega t) \end{aligned}$$

following expressions:

Recalling the geometric interpretation of the impedance Z

$$|Z| \cos \theta = R \quad \text{and} \quad |Z| \sin \theta = X$$

are the resistive and reactive components of the load impedance, respectively. On the basis of this fact, it becomes possible to write the instantaneous power as:

$$\begin{aligned} p(t) &= \tilde{I}^2 R (1 + \cos(2\omega t)) + \tilde{I}^2 X \sin(2\omega t) \\ &= \tilde{I}^2 R + \tilde{I}^2 R \cos(2\omega t) + \tilde{I}^2 X \sin(2\omega t) \end{aligned}$$

Since P_{av} corresponds to the power absorbed by the load resistance, it is also called the real power, measured in units of watts (W). On the other hand, Q takes the name of reactive power, since it is associated with the load reactance. The units of Q are volt-amperes reactive, or VAR. Note that Q represents an exchange of energy between the source and the reactive part of the load; thus, no net power is gained or lost in the process, since the average reactive power is zero. In general, it is desirable to minimize the reactive power in a load.

The computation of AC power is greatly simplified by defining a fictitious but very useful quantity called the complex power, S :

$$S = \tilde{\mathbf{V}} \tilde{\mathbf{I}}^*$$

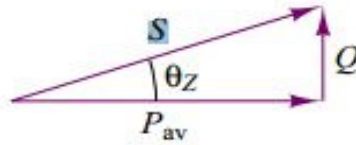
where the asterisk denotes the complex conjugate. You may easily verify that this definition leads to the convenient expression

$$S = \tilde{V} \tilde{I} \cos \theta + j \tilde{V} \tilde{I} \sin \theta = \tilde{I}^2 R + j \tilde{I}^2 X = \tilde{I}^2 Z$$

or

$$S = P_{av} + jQ$$

The complex power S may be interpreted graphically as a vector in the complex S plane



$$|S| = \sqrt{P_{av}^2 + Q^2} = \tilde{V} \cdot \tilde{I}$$

$$P_{av} = \tilde{V} \tilde{I} \cos \theta$$

$$Q = \tilde{V} \tilde{I} \sin \theta$$

The magnitude of S , $|S|$, is measured in units of volt-amperes (VA) and is called apparent power, because this is the quantity one would compute by measuring the rms load voltage and currents without regard for the phase angle of the load. The complex power may also be expressed by the product of the square of the rms current through the load and the complex load impedance:

$$S = \tilde{I}^2 Z$$

or

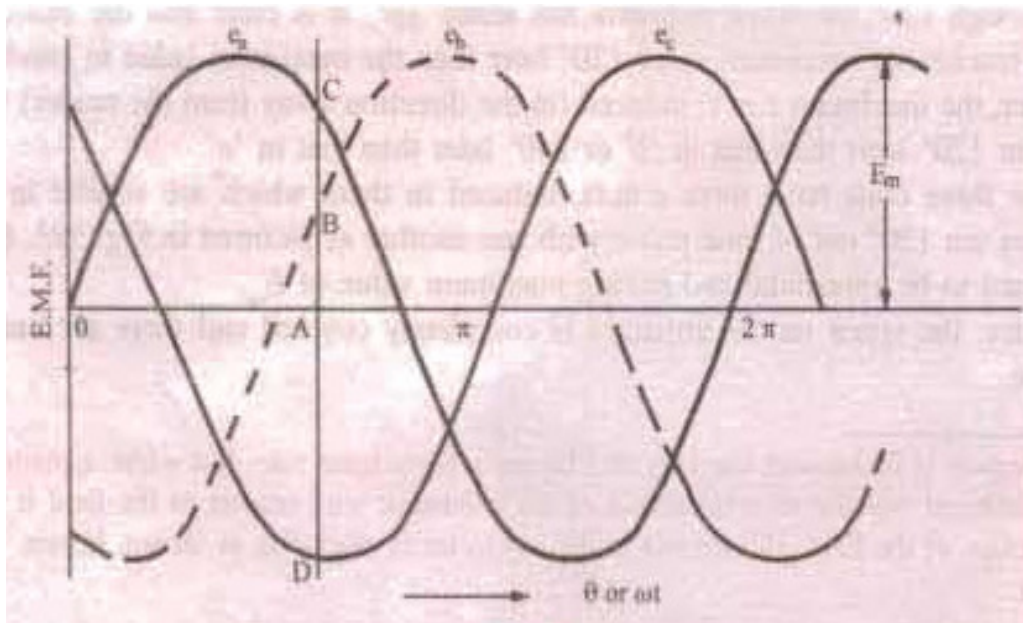
$$\tilde{I}^2 R + j \tilde{I}^2 X = \tilde{I}^2 Z$$

or, equivalently, by the ratio of the square of the rms voltage across the load to the complex conjugate of the load impedance:

$$S = \frac{\tilde{V}^2}{Z^*}$$

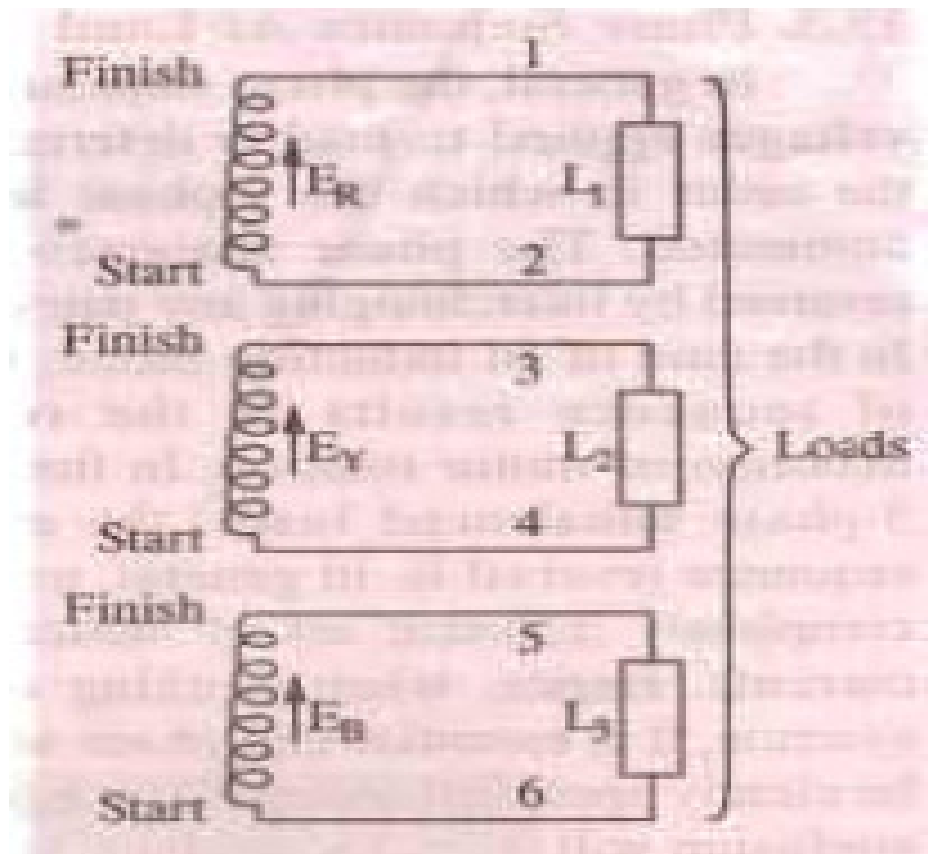
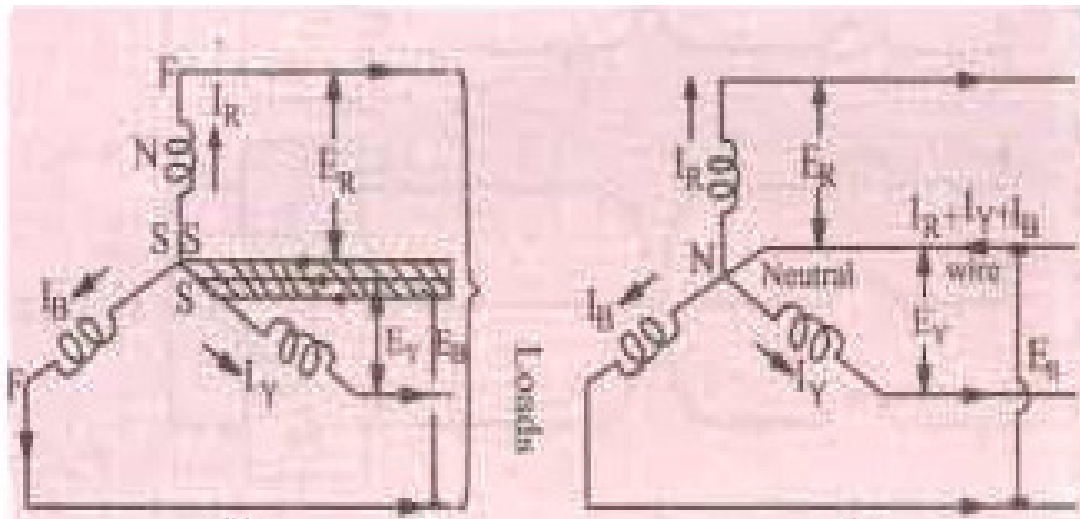
LECTURE-10
THREE PHASE POWER CALCULATIONS

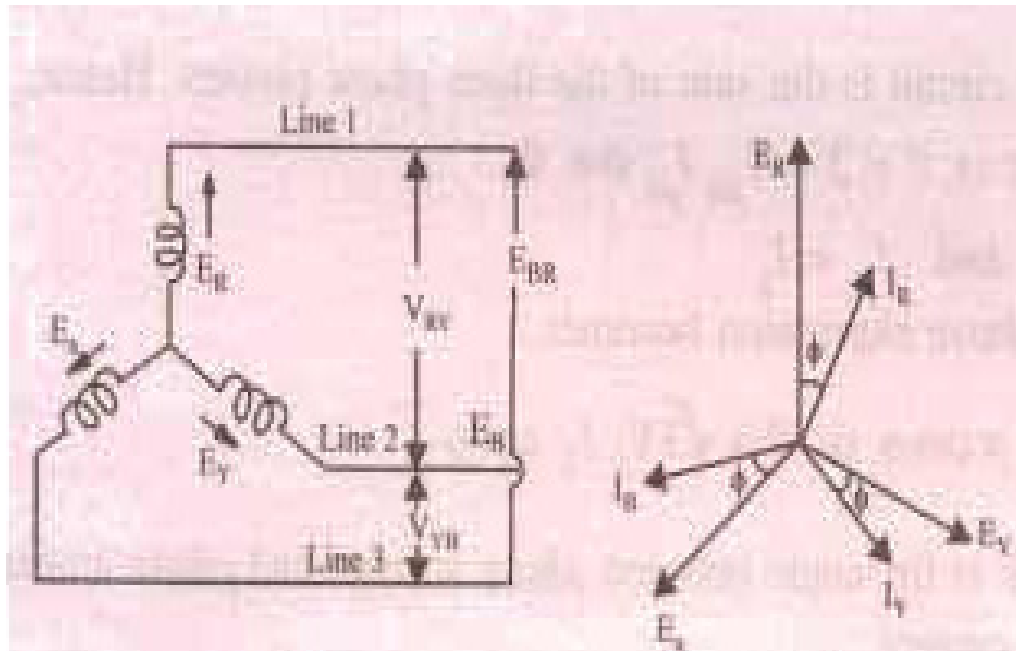
Three phase EMF Generation:-



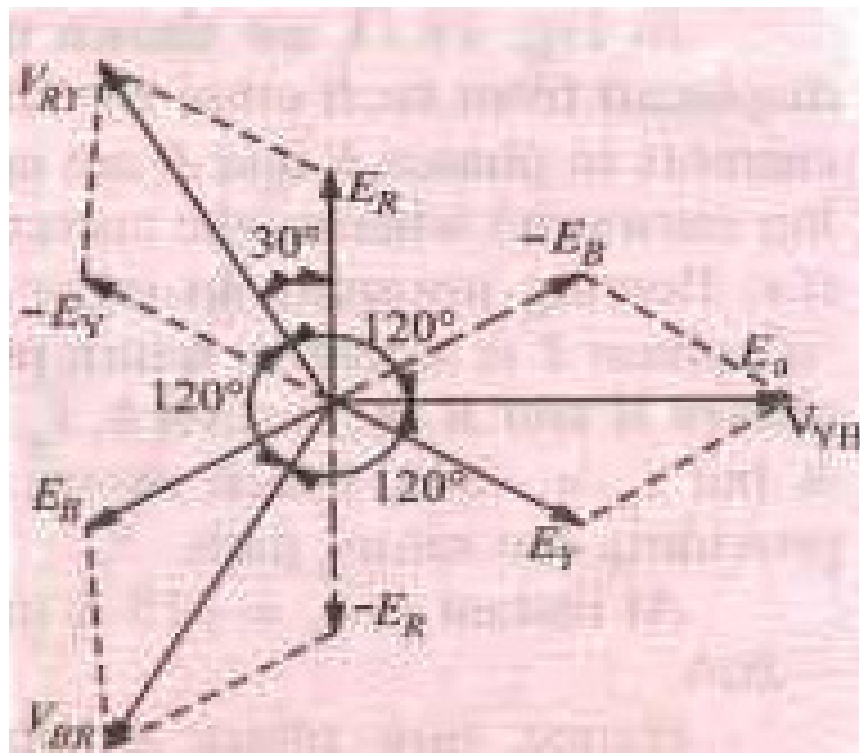
If the 3-coil windings W_1 , W_2 and W_3 arranged at 120° apart from each other on the same axis are rotated, then the emf induced in each of them will have a phase difference of 120° . In other words if the emf (or current) in one winding (w_1) has a phase of 0° , then the second winding (w_2) has a phase of 120° and the third (w_3) has a phase of 240° .

Star (Y) connection:-





Phasor diagram:-



Here, E_R , E_Y , E_B are phase voltages and V_{RY} , V_{YB} , V_{BR} are line voltages

$$\begin{aligned}
 V_{RY} &= \sqrt{E_R^2 + E_Y^2 + 2E_R E_Y \cos 60^\circ} \\
 &= \sqrt{E_R^2 + E_R^2 + 2E_R E_R \cos 60^\circ} \\
 &= \sqrt{3} E_R
 \end{aligned}$$

Hence,

- Line voltage $= \sqrt{3}$ x phase voltage
- Line current = phase current
- Line voltages are also 120° apart
- Line voltage are 30° ahead of respective phase voltages
- The angle between line voltage and line current is $(30^\circ + \Phi)$

Power: Total power = 3 x phase power

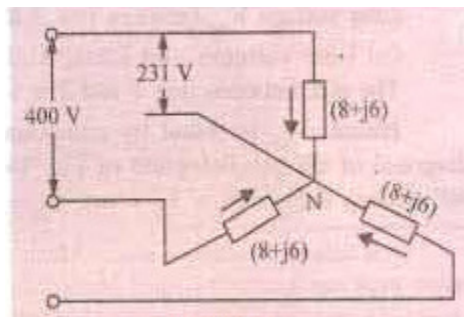
$$= 3 \times V_{ph} \times I_{ph} \times \cos \Phi$$

$$= \sqrt{3} V_L I_L \cos \Phi$$

Φ is the angle between phase voltage and current

Example: A balanced star connected load of $(8+j6)\Omega$ per phase is connected to a balanced 3-phase 400 V supply. Find the line current, power factor, power and total volt-amperes.

Ans.



$$Z_{ph} = \sqrt{8^2 + 6^2} = 10\Omega$$

$$V_{ph} = \frac{400}{\sqrt{3}} = 231\text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231}{10} = 23.1\text{ A}$$

$$(i) I_L = I_{ph} = 23.1\text{ A}$$

$$(ii) p.f = \cos\Phi = \frac{R_{ph}}{Z_{ph}} = \frac{8}{10} = 0.8(\text{lag})$$

$$(iii) \text{Power } P = \sqrt{3} V_L I_L \cos\Phi$$

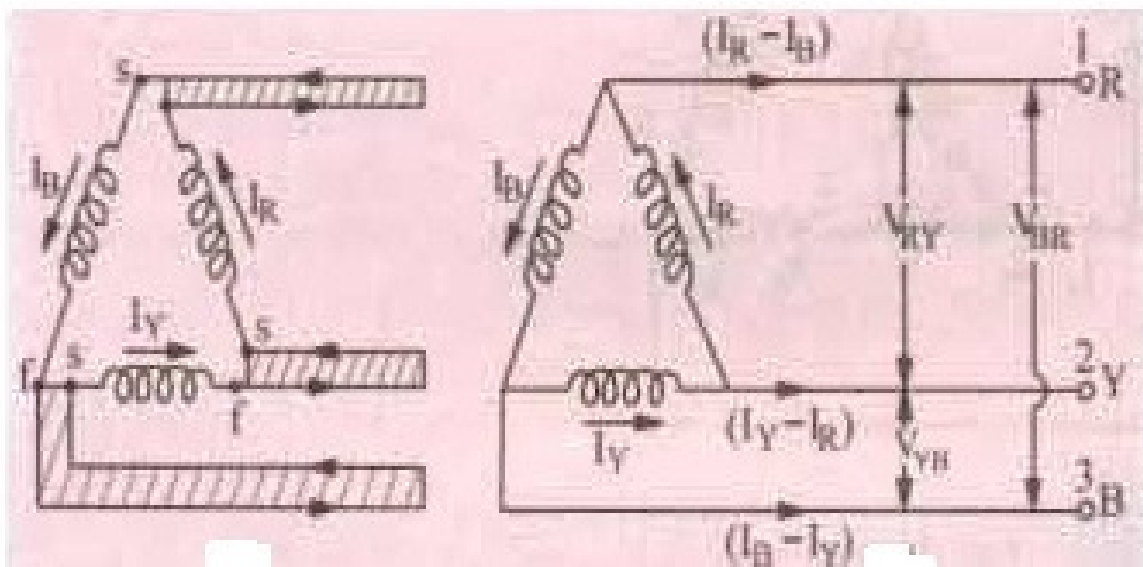
$$= \sqrt{3} \times 400 \times 23.1 \times 0.8$$

$$= 12,800\text{ W [Also, } P = 3I_{ph}^2 R_{ph} = 3(23.1)^2 \times 8 = 12,800\text{ W]}$$

$$(iv) \text{Total volt-amperes,}$$

$$S = \sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 23.1 = 16,000\text{ VA}$$

Delta-connection:



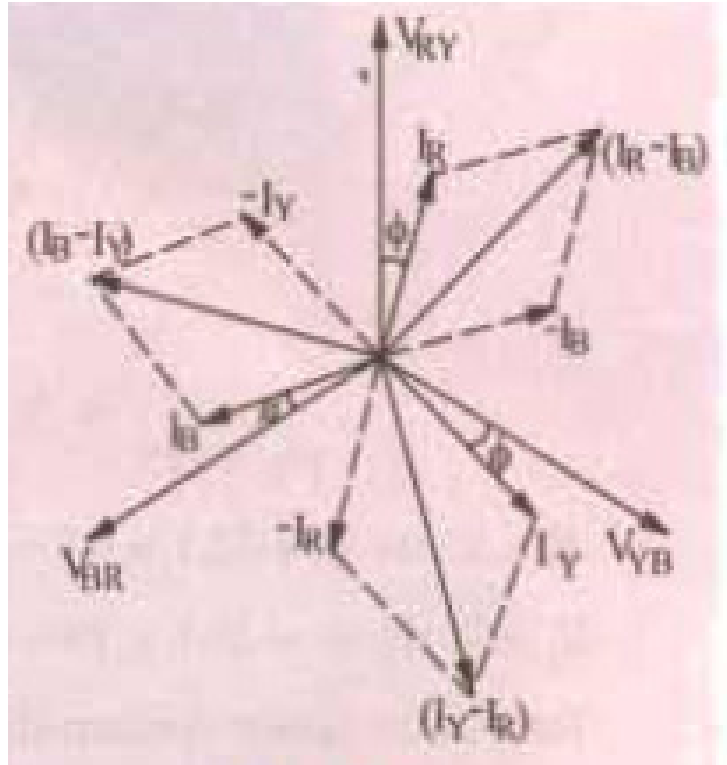


Fig. Phasor Diagram

$$I_L = I_R - I_B$$

$$I_L = \sqrt{I_R^2 + I_B^2 + 2I_R I_B \cos 60^\circ} = \sqrt{I_R^2 + I_R^2 + 2I_R I_R \cos 60^\circ} = \sqrt{3} I_R$$

Hence,

- Line current = $\sqrt{3}$ phase current
- Line voltage = phase voltage
- Line currents are also 120° apart
- Line currents are 30° behind the respective phase currents
- Angle between line current and line voltage is $30^\circ + \Phi$

Power: Total power = 3 x phase power

$$= 3 \times V_{ph} I_{ph} \cos \Phi$$

$$= 3 \times V_L \times I_L / \sqrt{3} \times \cos \Phi$$

$$= \sqrt{3} V_L I_L \cos \Phi$$

Note: For both star and delta system:

Active & True power = $\sqrt{3} V_L I_L \cos\Phi$

Reactive power = $\sqrt{3} V_L I_L \sin\Phi$

Apparent power = $\sqrt{3} V_L I_L$

LECTURE-11

SINGLE PHASE A.C CIRCUIT

Single phase EMF generation:

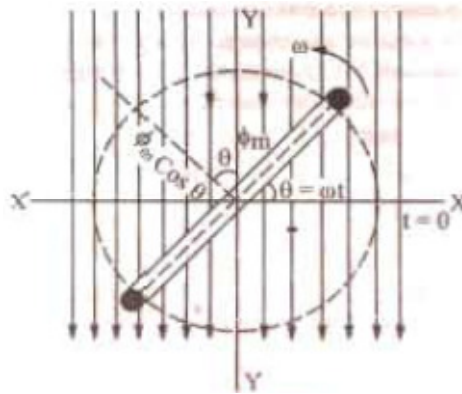
Alternating voltage may be generated

- 1) By rotating a coil in a magnetic field
- 2) By rotating a magnetic field within a stationary coil

The value of voltage generated depends upon

- 1) No. of turns in the coil
- 2) field strength
- 3) speed

Equation of alternating voltage and current



N = No. of turns in a coil

Φ_m = Maximum flux when coil coincides with X-axis

ω = angular speed (rad/sec) = $2\pi f$

At $\theta = \omega t$, Φ = flux component $\perp$ to the plane = $\Phi_m \cos \omega t$

According to the Faraday's law of electromagnetic induction,

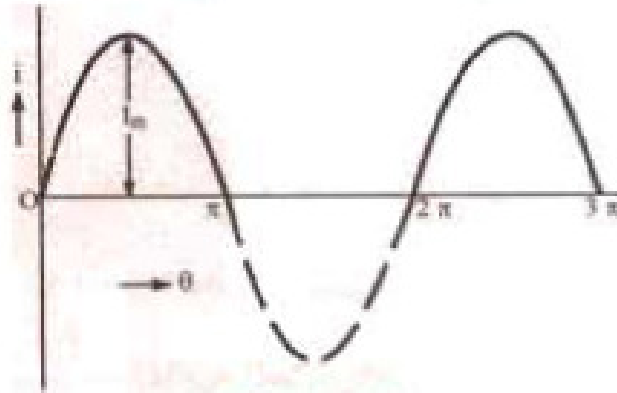
$$e = -N \frac{d\phi}{dt} = -N \frac{d}{dt} \phi_m \cos \omega t = \omega N \phi_m \sin \omega t \dots \dots \dots (1)$$

Now, e is maximum value of E_m , when $\sin \theta = \sin 90^\circ = 1$.

$$\text{i.e } E_m = \omega N \Phi_m \dots \dots \dots (2)$$

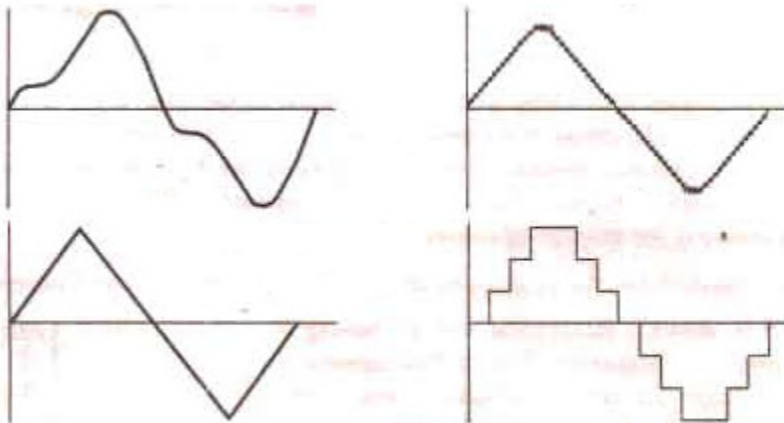
From Eqⁿ (1) & (2), $e = E_m \sin \omega t$ volt

Now, current (i) at any time in the coil is proportional to the induced emf (e) in the coil. Hence, $i = I_m \sin \omega t$ amp



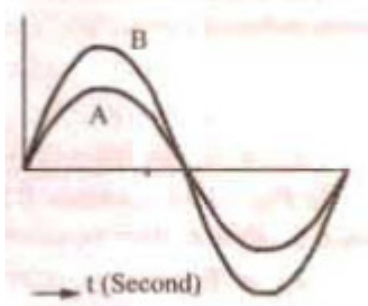
A.C terms:

- Cycle:- A complete set of positive and negative values of an alternating quantity is known as cycle.



- Time period: The time taken by an alternating quantity to complete one cycle is called time T.
- Frequency: It is the number of cycles that occur in one second. $f = 1/T$
 $f = PN/120$ where, P= No. of poles, N= Speed in rpm
- Waveform: A curve which shows the variation of voltage and current w.r.t time or rotation.

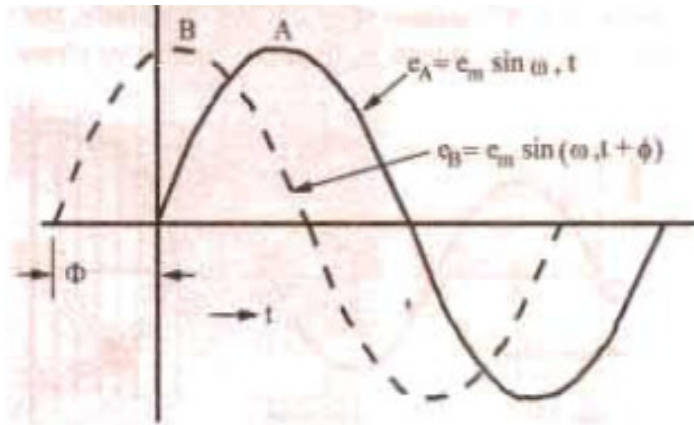
- Phase & Phase difference:



$$e_A = E_{mA} \sin \omega t$$

In phase: $e_B = E_{mB} \sin \omega t$

Out of phase: i) B leads A



$$e_A = E_m \sin \omega t$$

Phase difference Φ . $e_B = E_{mB} \sin (\omega t + \alpha)$

ii) A leads B or B lags A

$$e_A = E_m \sin \omega t$$

$$e_B = E_m \sin (\omega t - \alpha)$$

The vector I_{12} is reversed and is added in the vector I_{31} to get the vector sum of I_{31} and $-I_{12}$ as shown above in the phasor diagram. Therefore,

$$I_R = \sqrt{I_{31}^2 + I_{12}^2 + 2I_{31}I_{12} \cos 60^\circ} \text{ or}$$

$$I_L = \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}I_{ph} \times 0.5}$$

As we know, $I_R = I_L$, therefore,

$$I_L = \sqrt{3I_{ph}^2} = \sqrt{3}I_{ph}$$

Similarly,

$$\overline{I_Y} = \overline{I_{12}} - \overline{I_{23}} \text{ or } I_L = \sqrt{3}I_{ph} \text{ and}$$

$$\overline{I_B} = \overline{I_{23}} - \overline{I_{31}} \text{ or } I_L = \sqrt{3}I_{ph}$$

Hence, in Delta connection line current is root three times of phase current.

$$\text{Line Current} = \sqrt{3} \times \text{Phase Current}$$